

PRACTICAL - 10

AIM: Data Preparation and Transformation using Tableau Prep

- a) Connect and load data into Tableau Prep
- b) Clean and profile data
- c) Group and standardize values
- d) Merge mismatched fields
- e) Perform aggregations in Prep
- f) Apply filters and transformations
- g) Create and automate data preparation flows

A: Connect and Load Data into Tableau Prep

Objective: Establish connections to the local CSV data sources (Student_Assessments_Term1.csv, Student_Assessments_Term2.csv, and Course_Credit_Master.csv) and initiate a visual data preparation workflow.

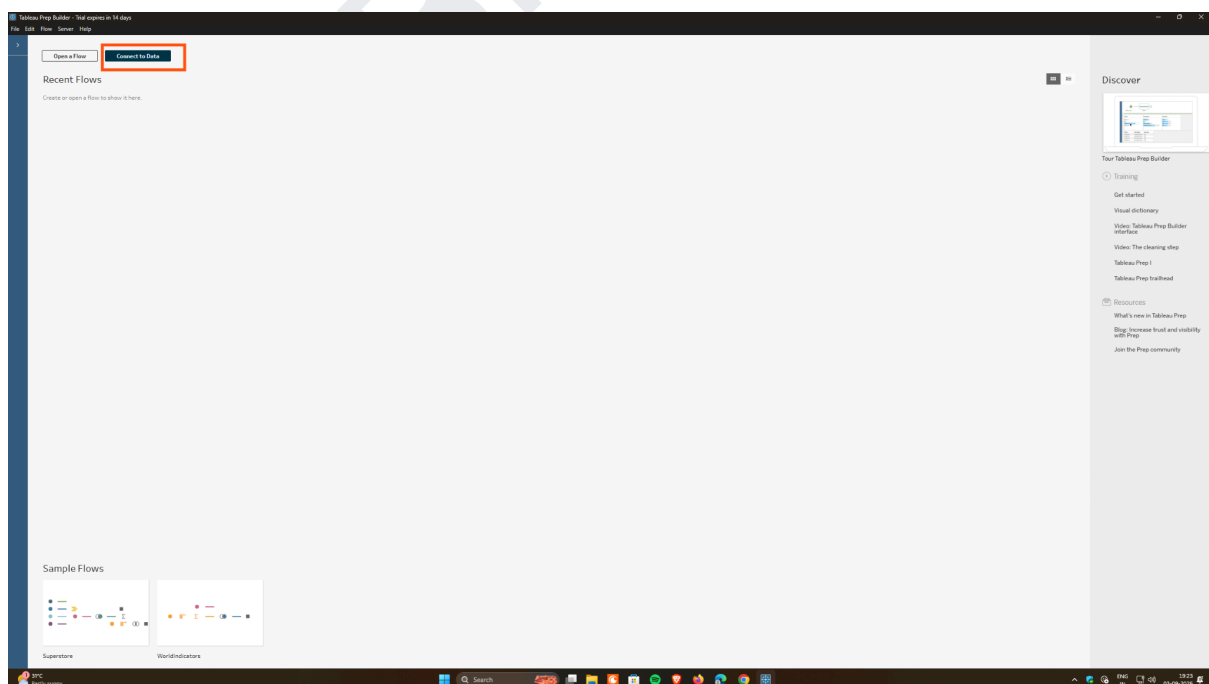
Tableau Prep Builder utilizes an input step pipeline architecture. Connecting multiple disparate files into a unified workspace establishes the foundation for schema validation, data type detection, and downstream data profiling.

Step 1: Open Tableau Prep Builder

1. Launch Tableau Prep Builder on your machine.
2. On the start screen, locate the Connections pane on the left-hand side.

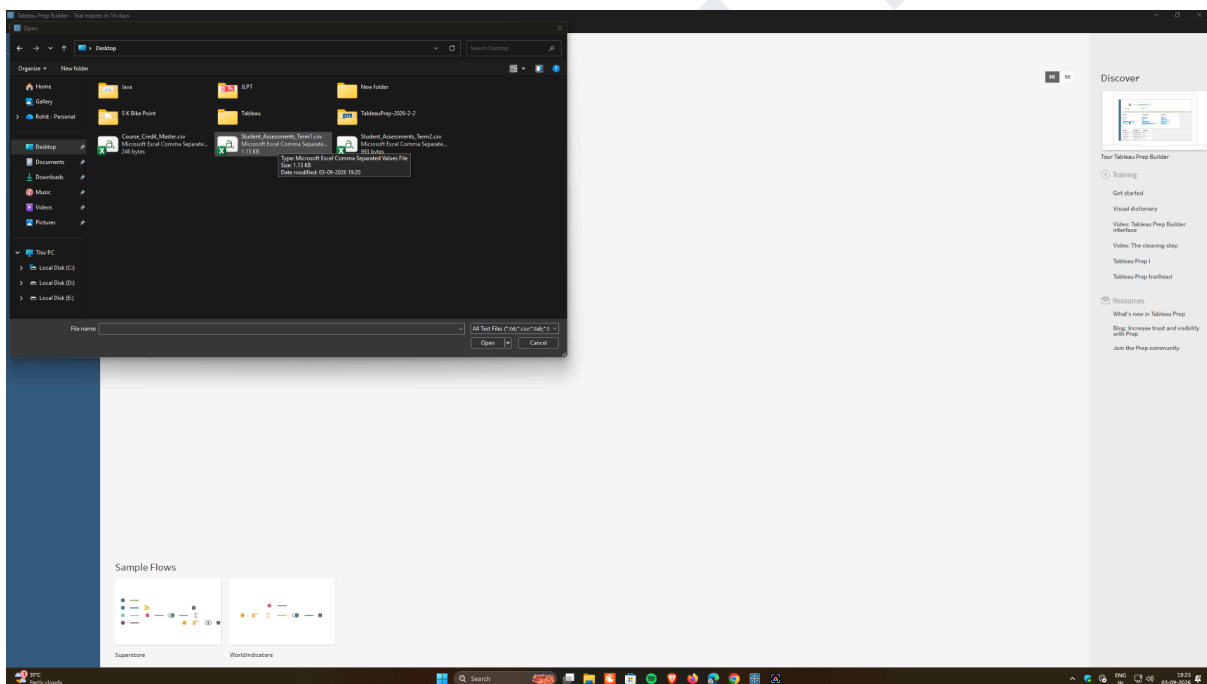
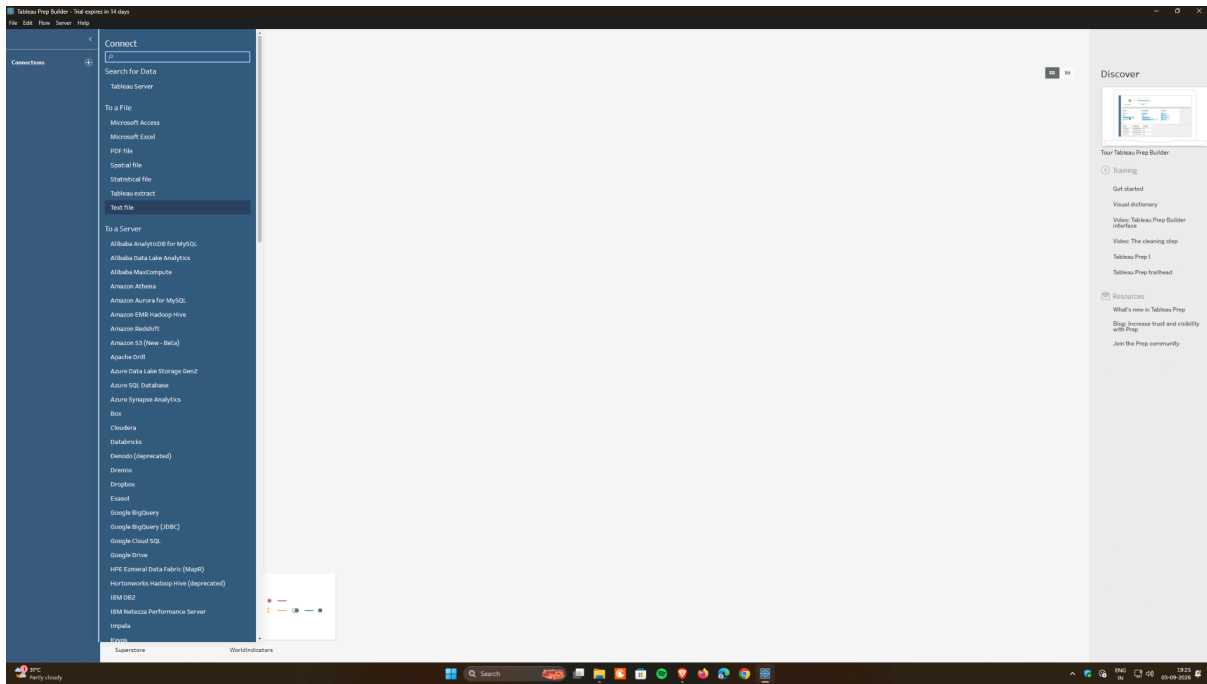
Step 2: Load the First Dataset (Student_Assessments_Term1.csv)

1. In the Connections pane, click the + (Add connection) button



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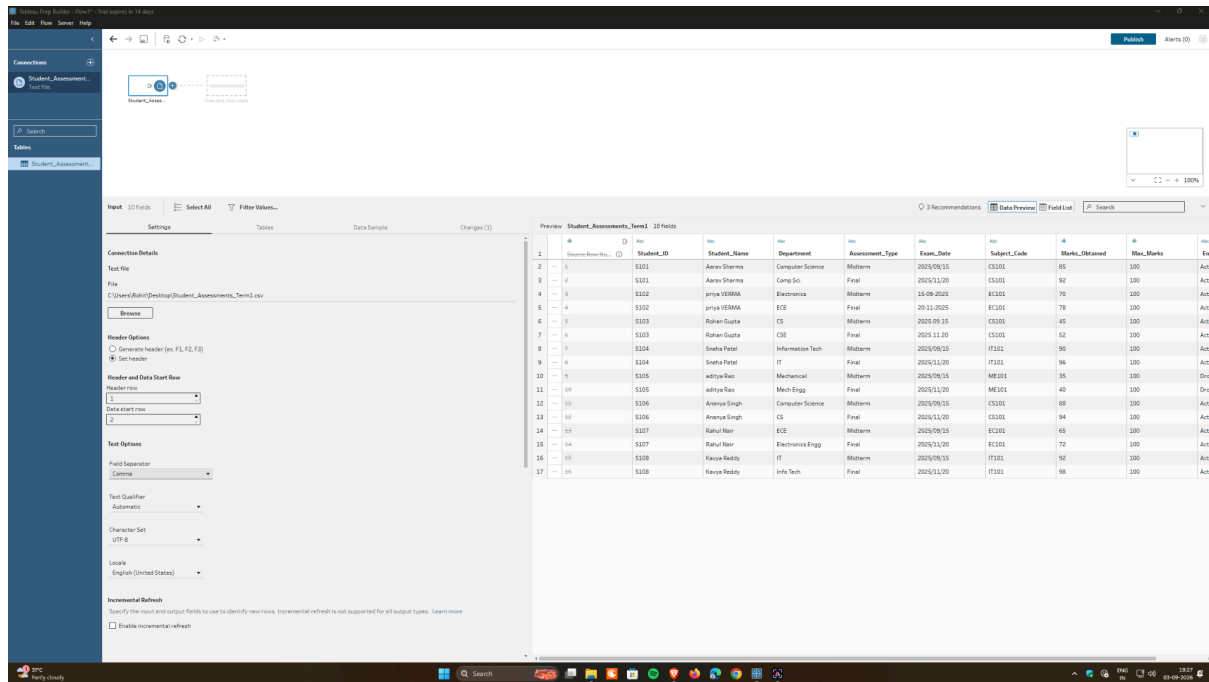
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2. In the list of file formats, select Text file.
3. Browse to the folder where you saved the CSVs, select Student_Assessments_Term1.csv, and click Open.
4. An Input Step icon named Student_Assessments_Term1 will appear on the Flow canvas.
5. In the bottom Input configuration pane, verify: o Field Separator: Comma (,)

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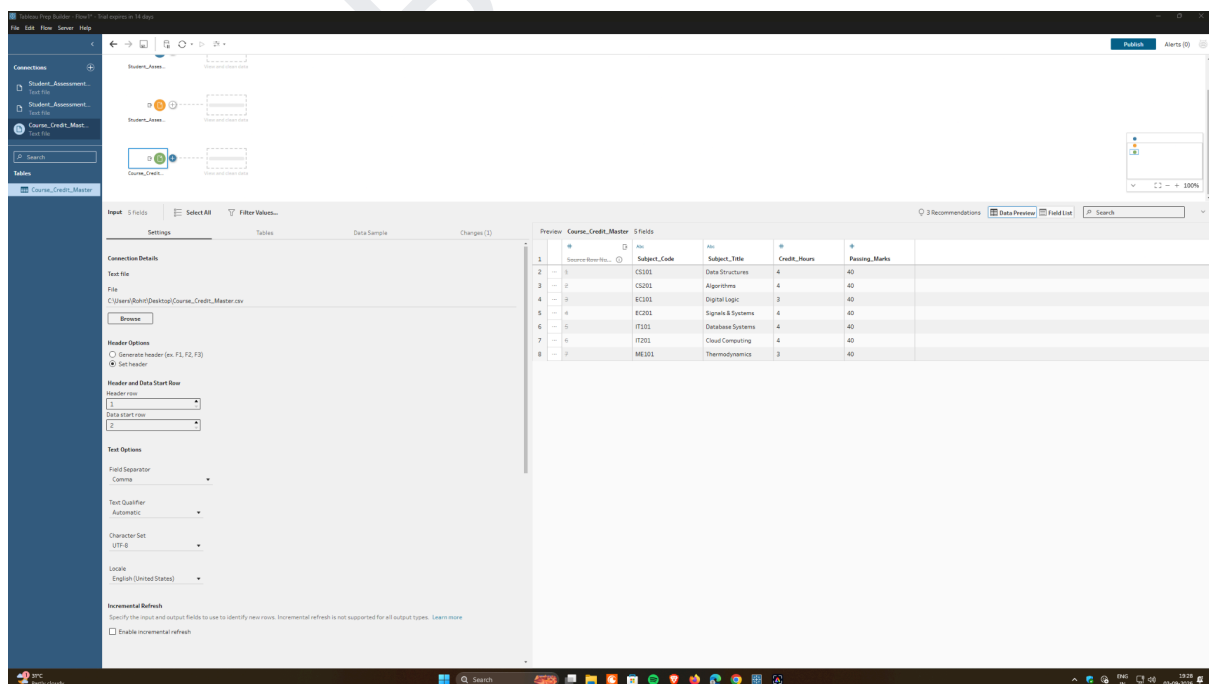
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- First row contains header: Checked

Step 3: Load the Second and Third Datasets

1. In the top-left Connections pane, click the + (Add connection) button again.
2. Select Text file, browse and select Student_Assessments_Term2.csv, and click Open.
3. Click + (Add connection) once more select Text file choose Course_Credit_Master.csv click Open.



4. You will now see three separate Input icons on your Flow canvas.

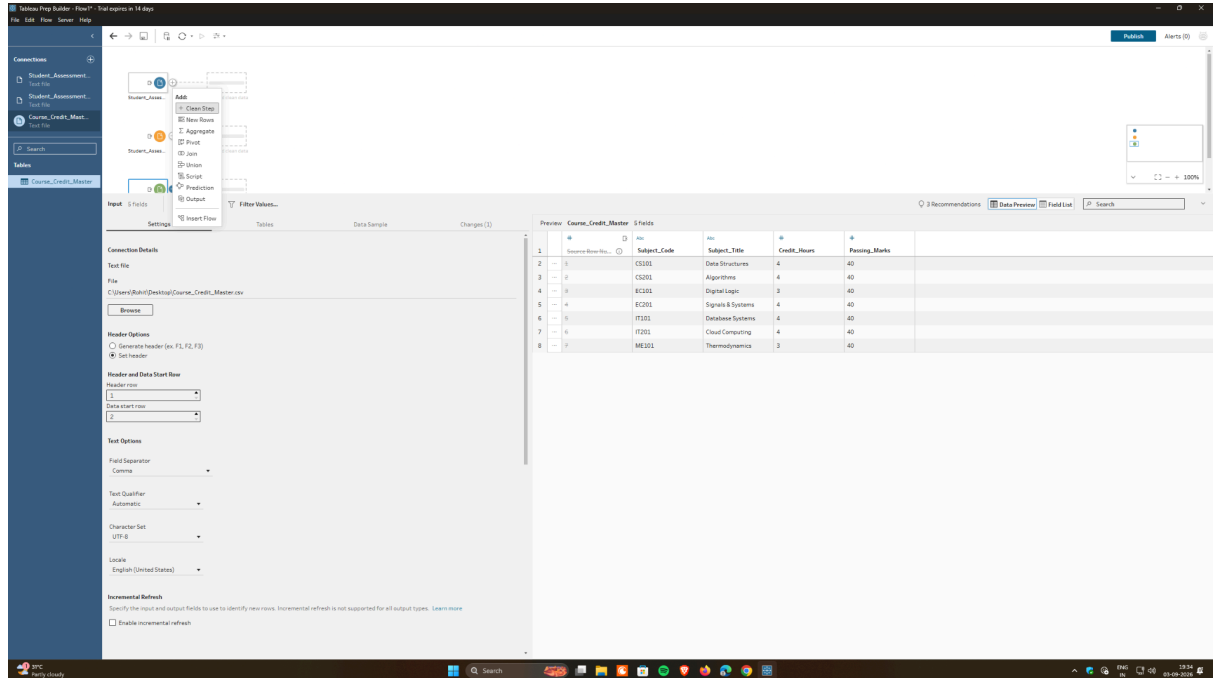
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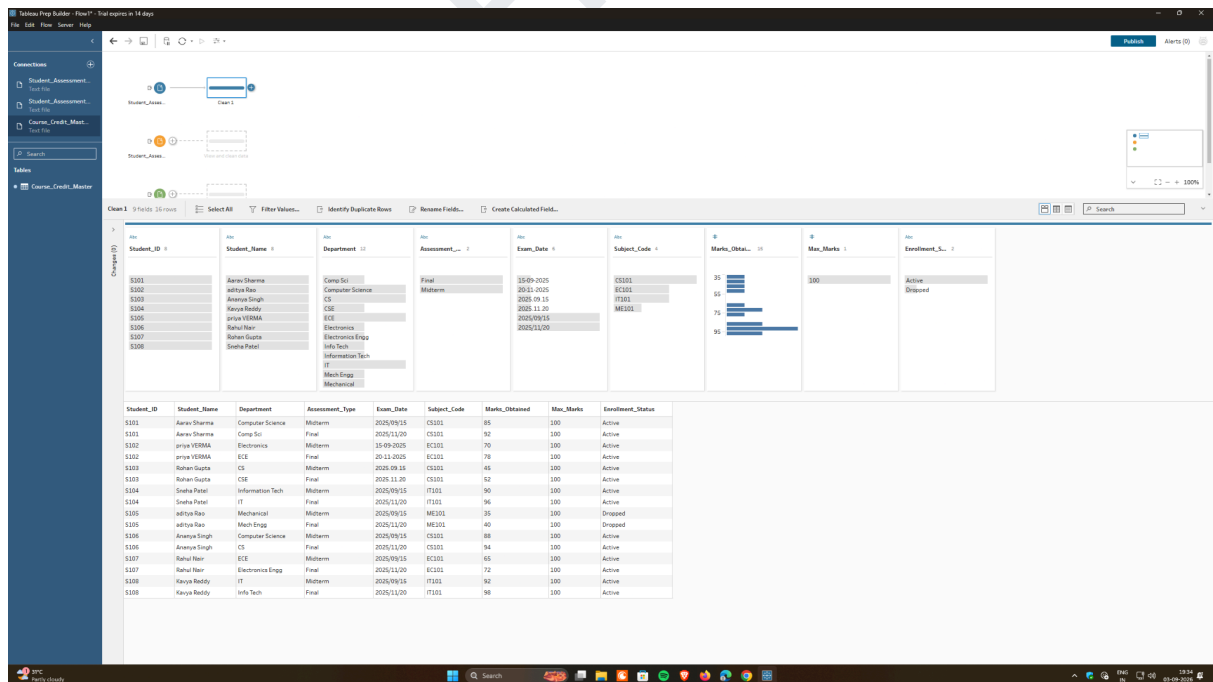
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Step 4: Add an Initial Clean Step to Term 1

1. On the Flow canvas, hover your mouse over the Student_Assessments_Term1 input icon.
2. Click the small + icon that appears to its right.
3. Select Clean Step (the step with the cleaning tool icon).



4. The Profile Pane will open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1.



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B: Clean and Profile Data

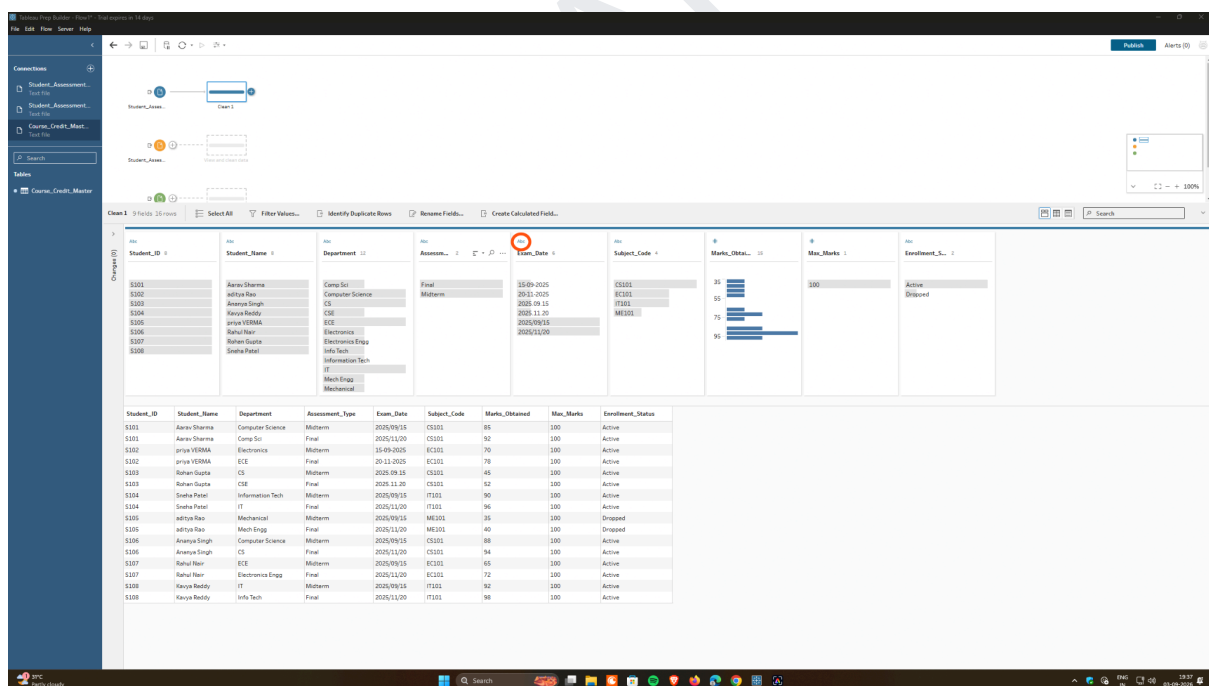
Objective: Use Tableau Prep Builder's interactive Profile Pane to identify data anomalies, whitespace issues, date formatting discrepancies, and inconsistent casing, then apply standard cleaning operations.

Unlike traditional spreadsheets, Tableau Prep provides real-time profiling cards that visualize distributions, frequency counts, and metadata data types for every field. Profiling allows you to quickly detect hidden formatting defects (such as extra spaces in names like 'Aarav Sharma ' or non-standard date strings) before standardizing categorical groups or performing joins.

Step 1: Inspect the Profile Pane

1. Click on the Clean 1 step connected to Student_Assessments_Term1. 2. Look at the cards displayed in the Profile Pane:

- o Each card represents one column.
- o Notice that Student_Name has multiple representations of the same student due to extra spaces and mixed casing (e.g., 'Aarav Sharma ' vs 'Aarav Sharma', 'priya VERMA', 'aditya Rao ').
- o Notice that Exam_Date is recognized as a string (Abc) due to mixed separators (/, -, .).



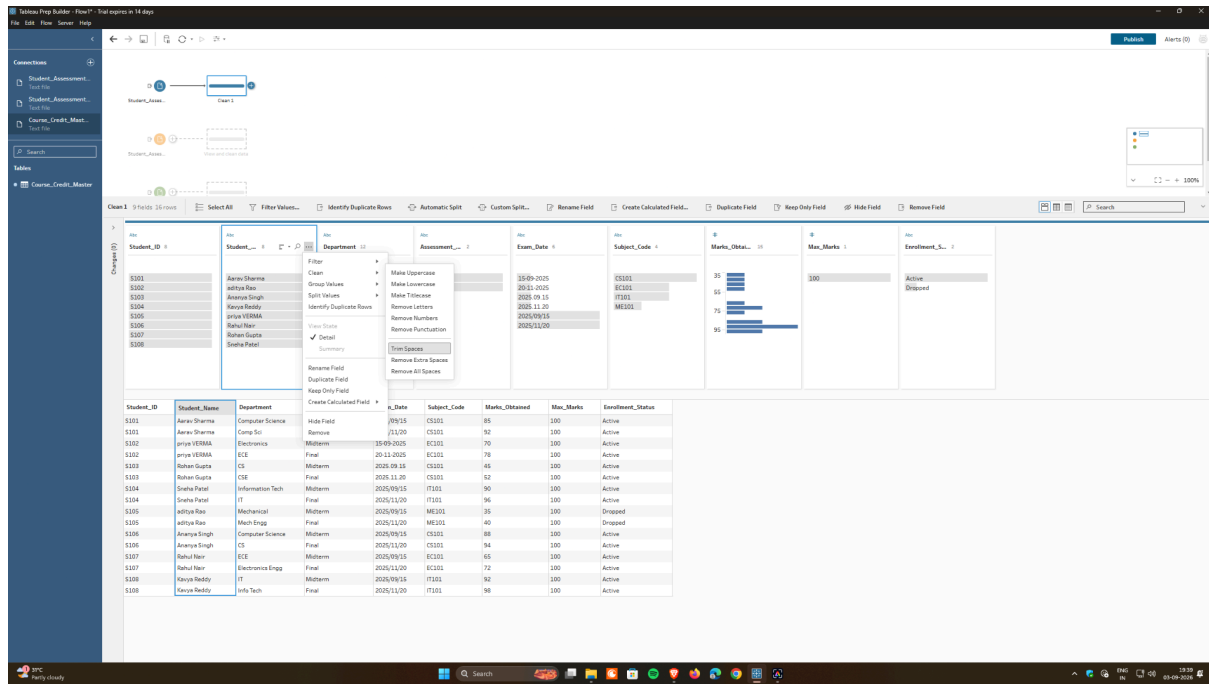
Step 2: Clean Whitespace and Standardize Letter Casing

1. On the Student_Name profile card:

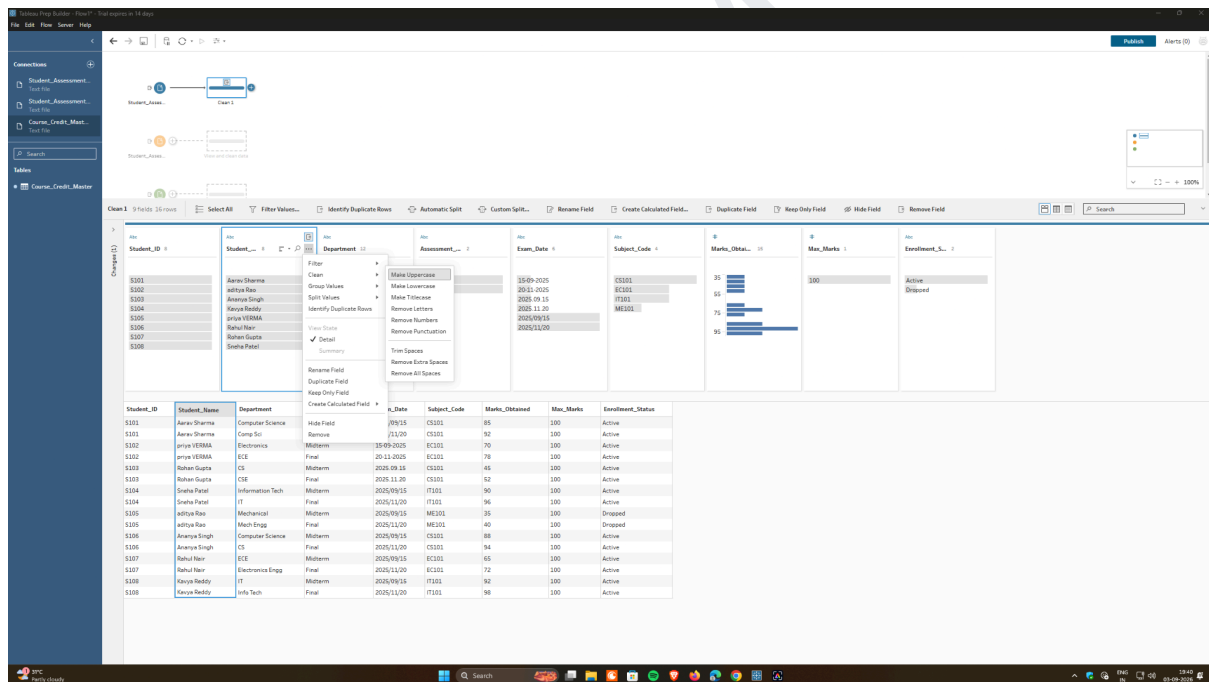
- o Click the More options (...) icon at the top right of the card.
- o Go to Clean --> select Trim Spaces (this strips all leading and trailing whitespace).

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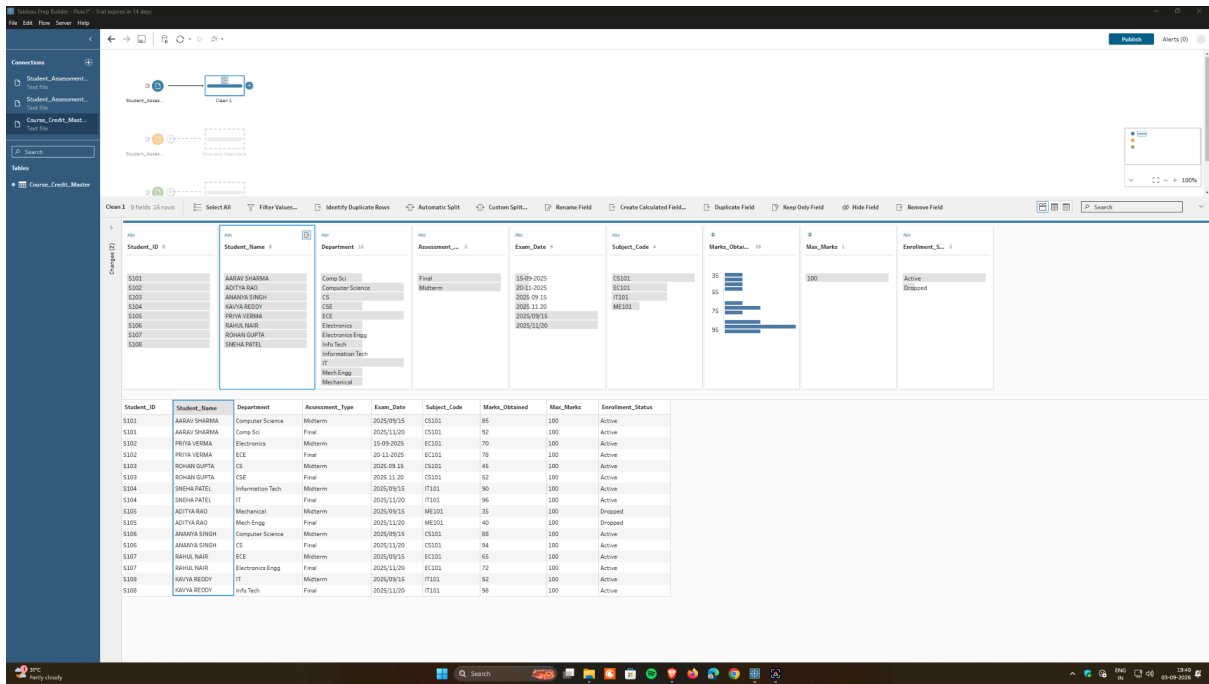
o Click More options (...) again --> Clean --> select Make Uppercase.



2. The card updates immediately, consolidating duplicate whitespace variations.

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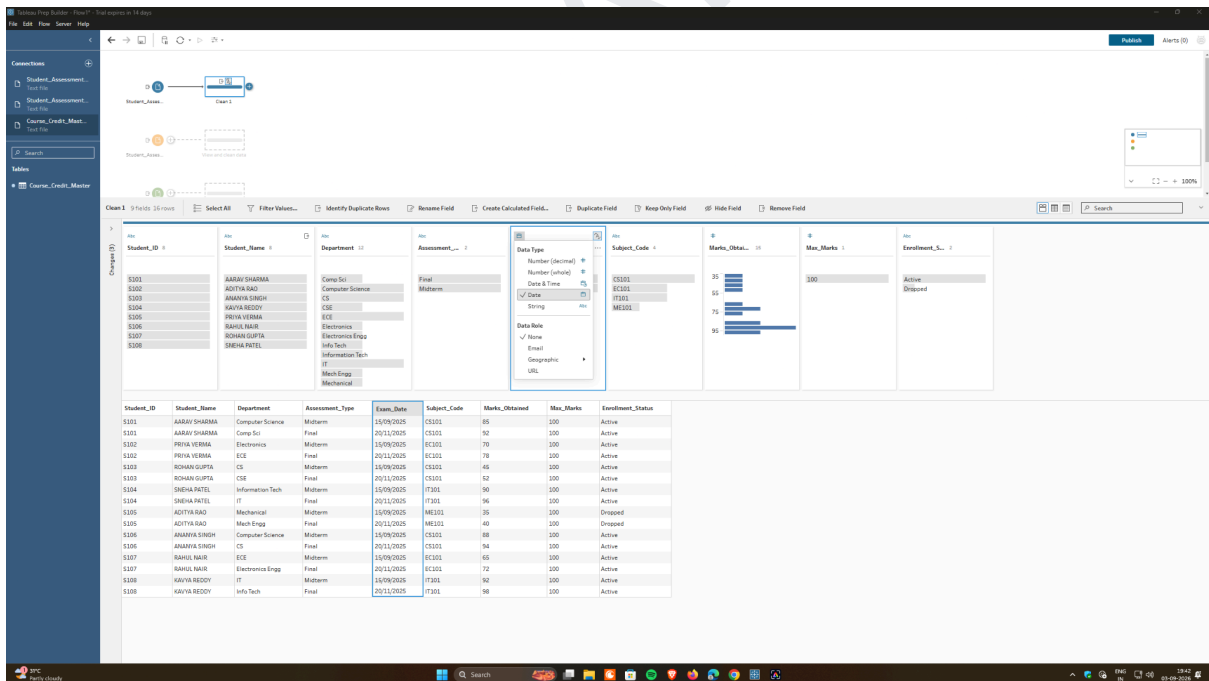
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Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S101	AARAV SHARMA	Computer Science	Midterm	2025/09/15	CS101	85	100	Active
S101	AARAV SHARMA	Comp Sci	Final	2025/11/20	CS101	92	100	Active
S102	PRINIA VERMA	Electronics	Midterm	15-09-2025	EC101	78	100	Active
S102	PRINIA VERMA	ECE	Final	20-11-2025	EC101	79	100	Active
S103	ROHAN GUPTA	CS	Midterm	2025/09/15	CS101	45	100	Active
S103	ROHAN GUPTA	CSE	Final	2025/11/20	CS101	52	100	Active
S104	SNEHA PATEL	Information Tech	Midterm	2025/09/15	IT101	90	100	Active
S104	SNEHA PATEL	IT	Final	2025/11/20	IT101	96	100	Active
S105	ADITYA RAO	Mechanical	Midterm	2025/09/15	ME101	35	100	Dropped
S105	ADITYA RAO	Mech Engg	Final	2025/11/20	ME101	40	100	Dropped
S106	ANANYA SINGH	Computer Science	Midterm	2025/09/15	CS101	88	100	Active
S106	ANANYA SINGH	CS	Final	2025/11/20	CS101	94	100	Active
S107	RAHUL NAR	ECE	Midterm	2025/09/15	EC101	65	100	Active
S107	RAHUL NAR	Electronics Engg	Final	2025/11/20	EC101	72	100	Active
S108	KAVYA REDDY	IT	Midterm	2025/09/15	IT101	92	100	Active
S108	KAVYA REDDY	Info Tech	Final	2025/11/20	IT101	98	100	Active

Step 3: Convert Exam_Date to Date Data Type

1. Locate the Exam_Date profile card.
2. Click the data type icon (currently showing Abc) directly above the column name.
3. Select Date from the dropdown menu.

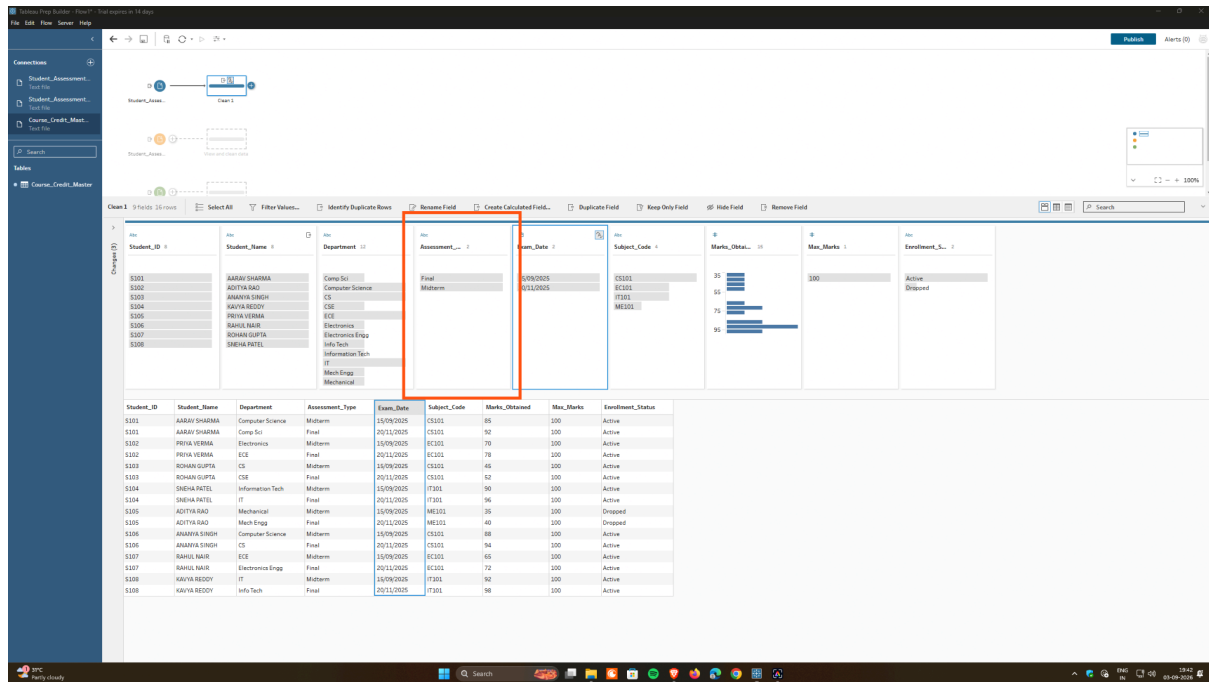


Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S101	AARAV SHARMA	Computer Science	Midterm	15/09/2025	CS101	85	100	Active
S101	AARAV SHARMA	Comp Sci	Final	20/11/2025	CS101	92	100	Active
S102	PRINIA VERMA	Electronics	Midterm	15/09/2025	EC101	78	100	Active
S102	PRINIA VERMA	ECE	Final	20/11/2025	EC101	79	100	Active
S103	ROHAN GUPTA	CS	Midterm	15/09/2025	CS101	45	100	Active
S103	ROHAN GUPTA	CSE	Final	20/11/2025	CS101	52	100	Active
S104	SNEHA PATEL	Information Tech	Midterm	15/09/2025	IT101	90	100	Active
S104	SNEHA PATEL	IT	Final	20/11/2025	IT101	96	100	Active
S105	ADITYA RAO	Mechanical	Midterm	15/09/2025	ME101	35	100	Dropped
S105	ADITYA RAO	Mech Engg	Final	20/11/2025	ME101	40	100	Dropped
S106	ANANYA SINGH	Computer Science	Midterm	15/09/2025	CS101	88	100	Active
S106	ANANYA SINGH	CS	Final	20/11/2025	CS101	94	100	Active
S107	RAHUL NAR	ECE	Midterm	15/09/2025	EC101	65	100	Active
S107	RAHUL NAR	Electronics Engg	Final	20/11/2025	EC101	72	100	Active
S108	KAVYA REDDY	IT	Midterm	15/09/2025	IT101	92	100	Active
S108	KAVYA REDDY	Info Tech	Final	20/11/2025	IT101	98	100	Active

4. Tableau Prep automatically parses the various date formats (2025/09/15, 15-09-2025, 2025.11.20) into standardized calendar dates.

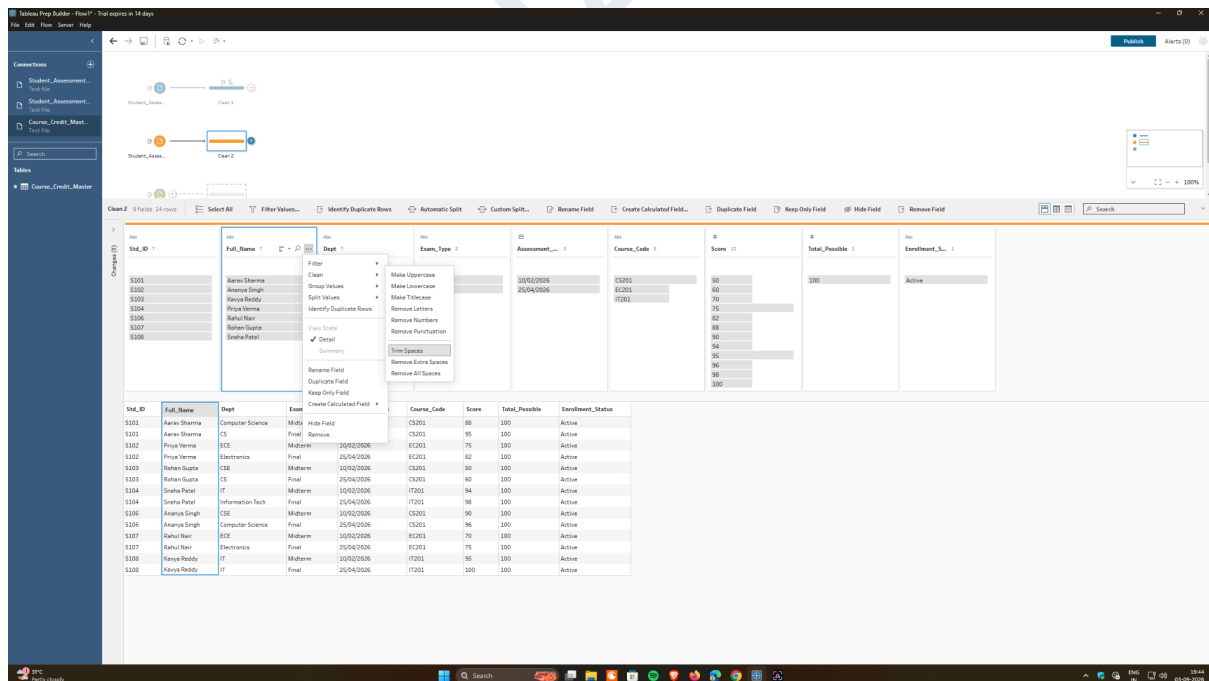
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Step 4: Clean the Second Dataset (Student_Assessments_Term2.csv)

1. On the Flow canvas, hover over the Student_Assessments_Term2 input icon.
2. Click + --> select Clean Step (creates Clean 2).
3. On the Full_Name card in Clean 2: o Click More options (...) --> Clean --> Trim Spaces.

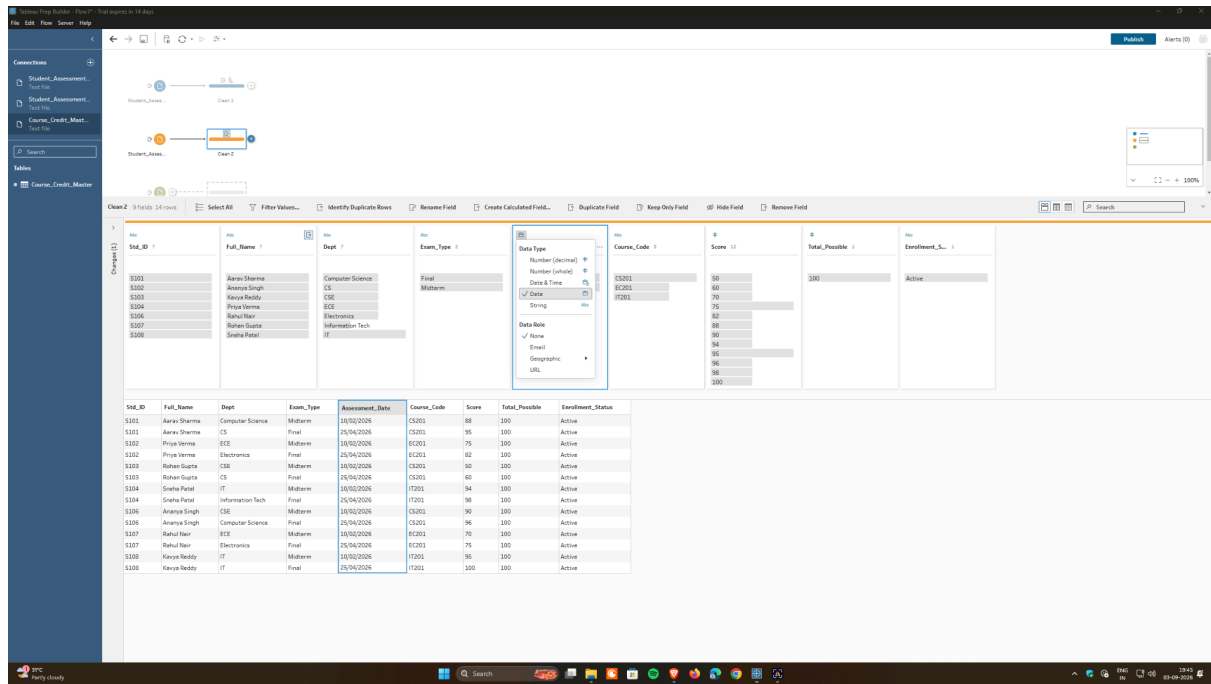


4. On the Assessment_Date card: o Click the Abc icon --> change the data type to Date.

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Step 5: Review the Changes Pane

1. Look at the left side of the screen under the Changes tab.
2. Notice that Tableau Prep records every single transformation as an itemized, reversible step (e.g., Trim spaces on Student_Name, Change data type of Exam_Date).

Profile & Cleaning Audit Table

Field Name	Detected Issue	Cleaning Operation Applied	Target State
Student_Name	Leading/trailing spaces (' Aarav Sharma ')	Clean Trim Spaces	Single trimaligned name
Student_Name	Inconsistent casing ('priya VERMA')	Clean Make Uppercase	Standardized casing
Exam_Date	Text datatype with mixed separators	Change Data Type Date	Standard Date format (YYYYMM-DD)
Assessment_Date	String data type	Change Data Type Date	Standard Date format (YYYYMM-DD)

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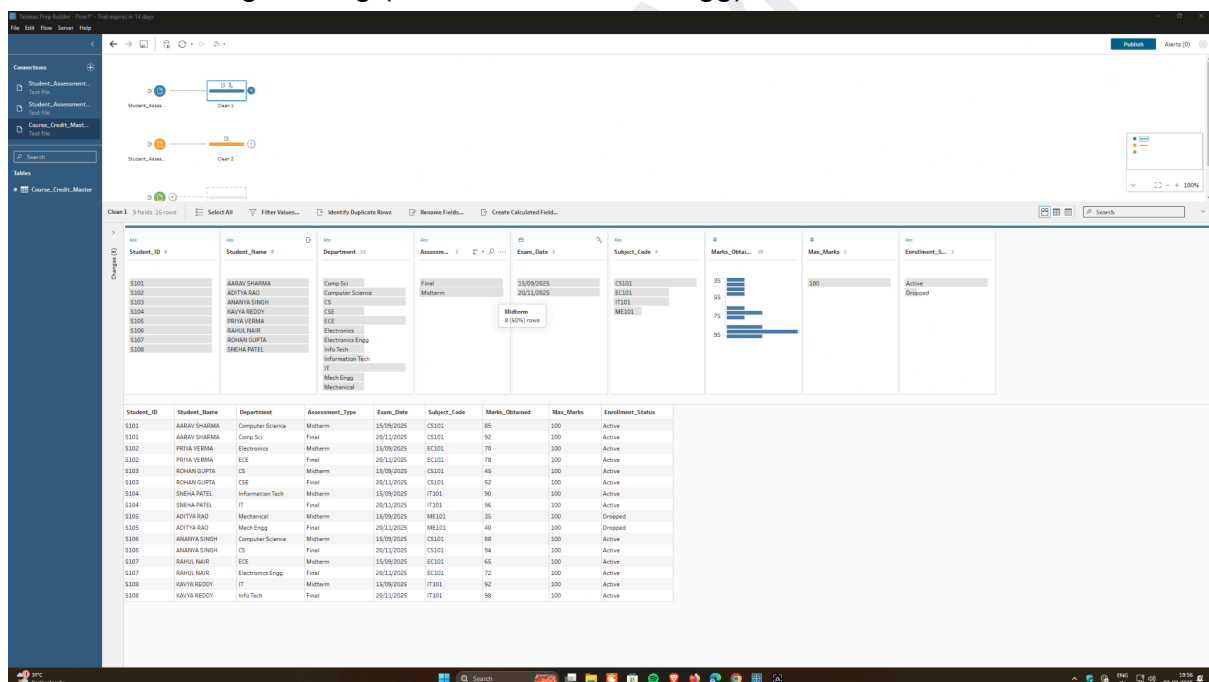
C: Group and Standardize Values

Objective: Standardize spelling variations, abbreviations, and synonyms in categorical columns (such as Department and Dept) into clean, consistent values using Tableau Prep's built-in grouping algorithms.

In academic datasets, departments are frequently entered under different aliases (e.g., "Comp Sci", "CS", "CSE", and "Computer Science"). If left unstandardized, downstream visual analysis will treat each alias as a completely different academic department, splintering the summary metrics.

Step 1: Inspect the Department Field in Term 1 (Clean 1)

1. Click on the Clean 1 step connected to Student_Assessments_Term1.
2. Locate the Department card in the Profile Pane.
3. Observe that there are 12 distinct entries representing only 4 actual academic departments:
 - o Computer Science (Computer Science, Comp Sci, CS, CSE)
 - o Electronics (Electronics, ECE, Electronics Engg)
 - o Information Technology (Information Tech, IT, Info Tech)
 - o Mechanical Engineering (Mechanical, Mech Engg)

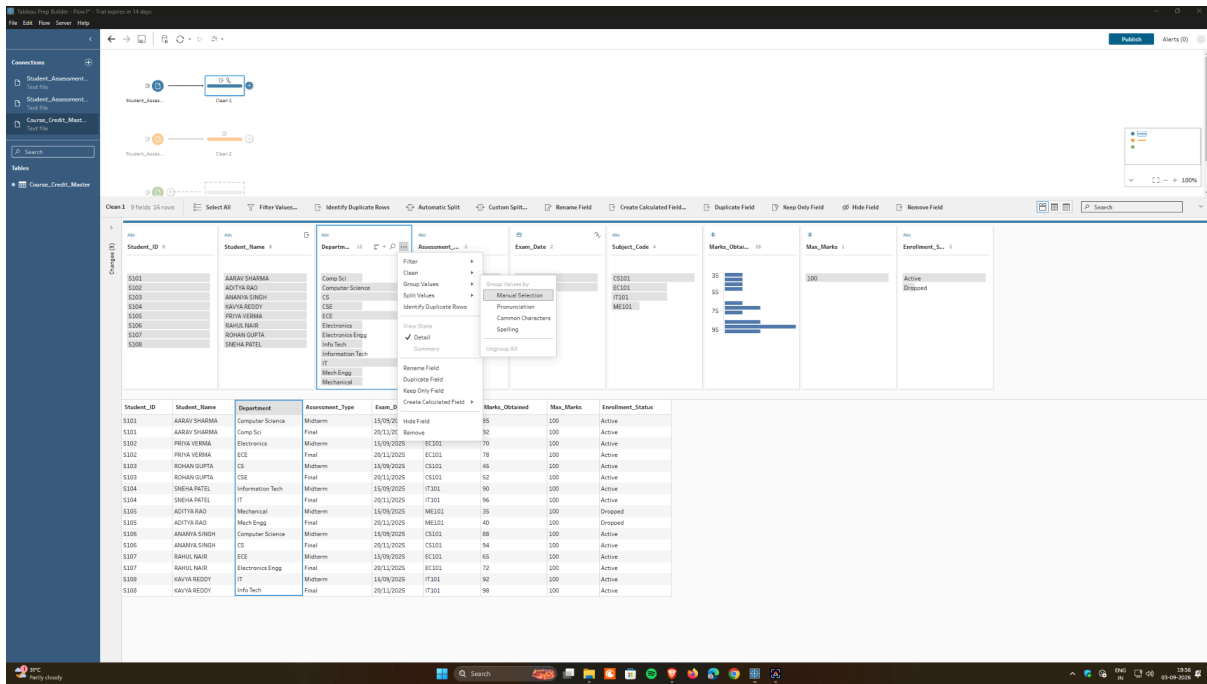


Step 2: Group Values Using Pronunciation / Spelling or Manual Selection

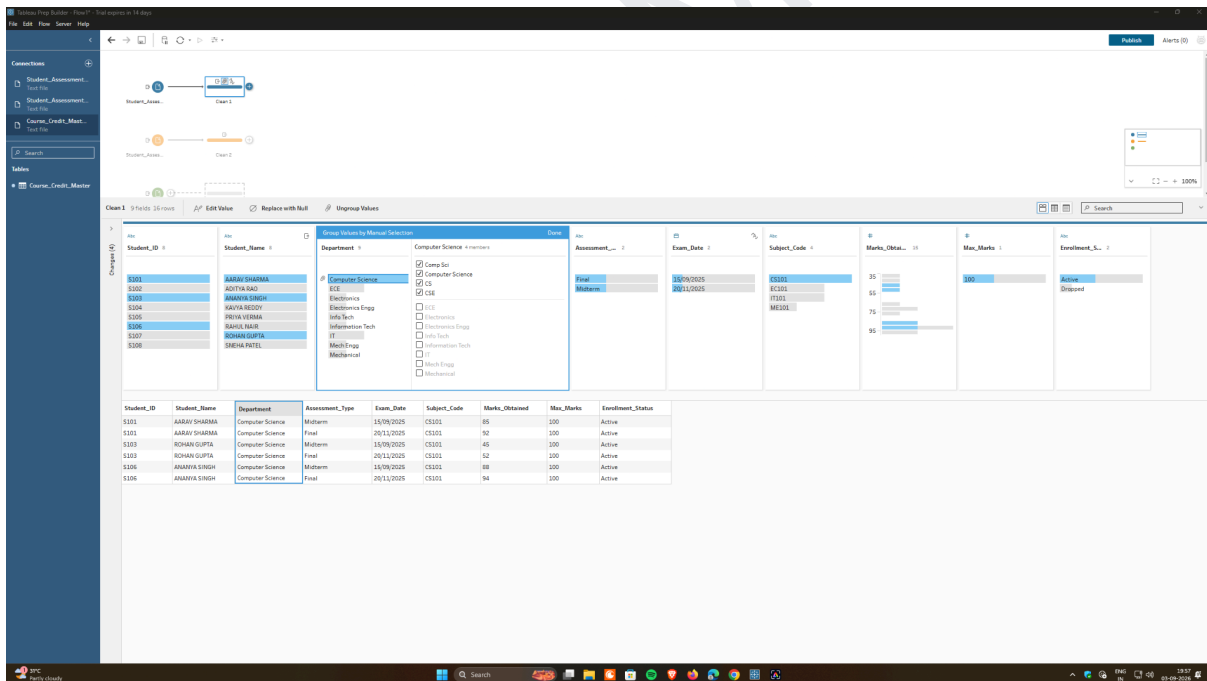
1. On the Department card, click the More options (...) menu at the top right.
2. Go to Group Values --> select Manual Selection (or Pronunciation / Common Characters).

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3. In the Grouping editor: o Click on Computer Science in the left column. In the right-hand panel, check the boxes for Comp Sci, CS, and CSE.



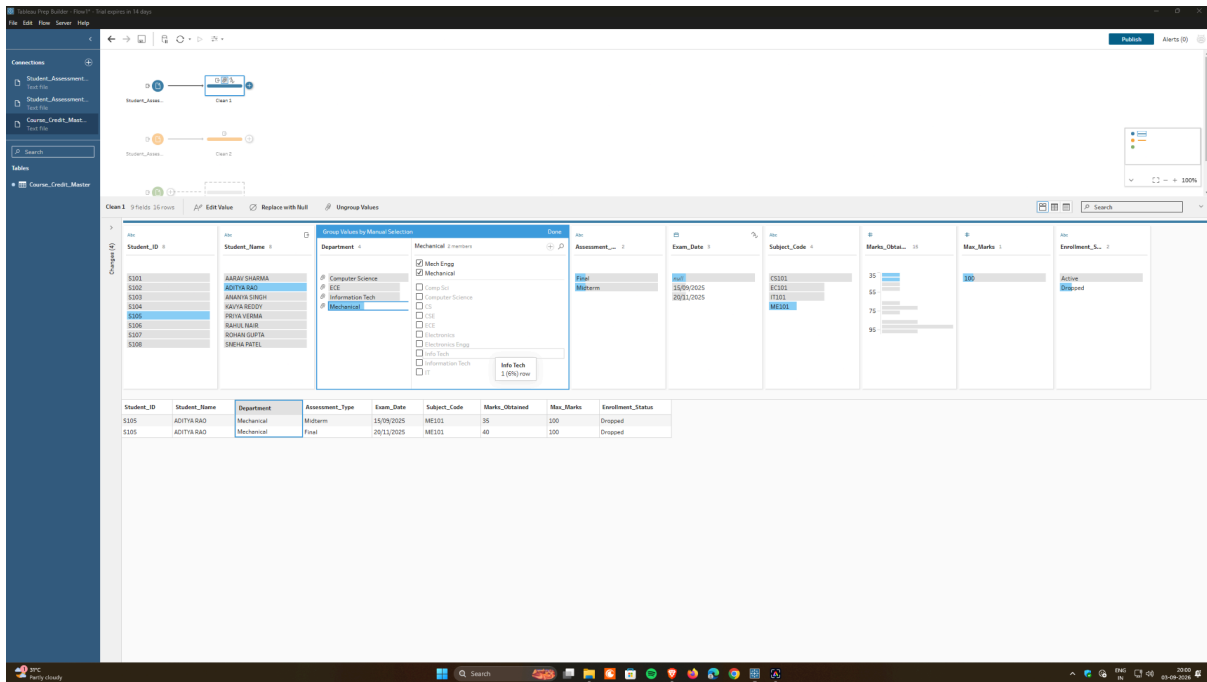
- Click on Electronics (or rename the master value to Electronics & Communication). Check the boxes for ECE and Electronics Engg.

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- The screenshot shows the Power BI Desktop interface with a data model and a dashboard. The data model includes tables like Student_Assessment, Course_Credit_Master, and Group_Venue. The dashboard displays a table of student marks and a bar chart of marks obtained.

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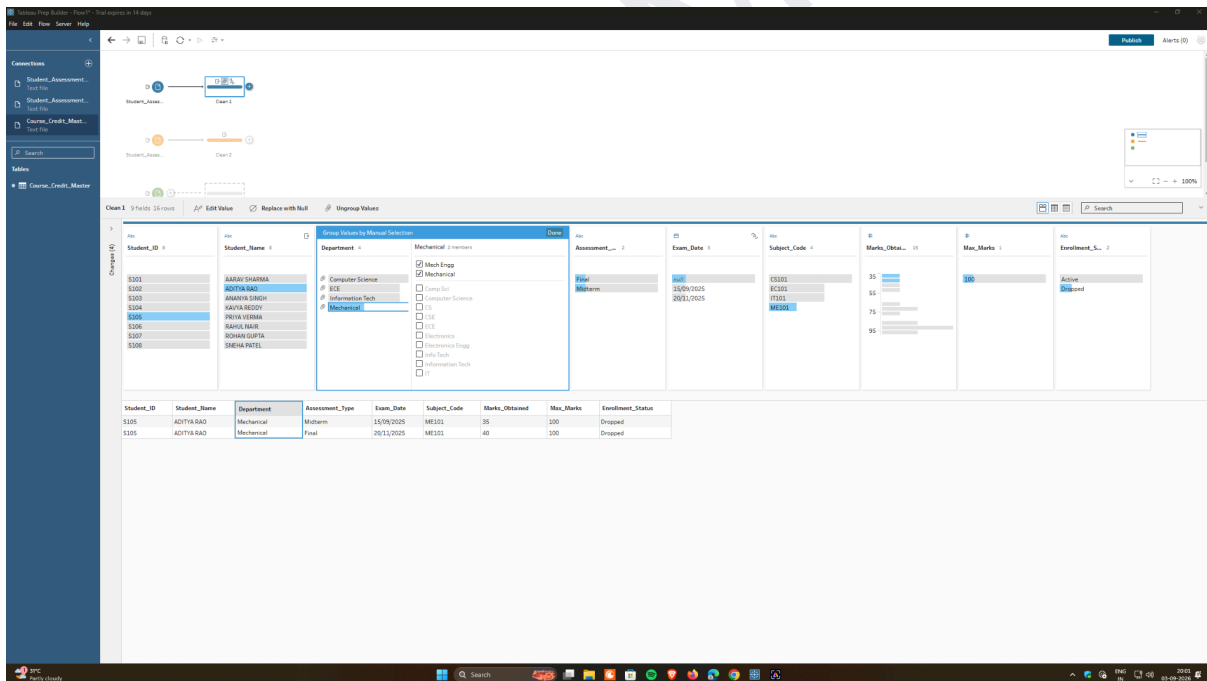
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4. Click Done at the top right of the editor

5. Notice that the paperclip icon appears next to grouped values, consolidating the 12 messy entries into exactly 4 clean departments.



Step 3: Repeat Grouping on Term 2 (Clean 2)

1. Click on the Clean 2 step connected to Student_Assessments_Term2.
2. Locate the Dept card.
3. Click More options (...) -->Group Values -->Manual Selection:
 - o Group CS, CSE, and Computer Science into Computer Science.
 - o Group ECE and Electronics into Electronics.
 - o Group IT and Information Tech into Information Tech.

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4. Click Done.

Standardization Mapping Table

Original Input Variations	Target Standardized Value	Department Code
Computer Science, Comp Sci, CS, CSE	Computer Science	CS
Electronics, ECE, Electronics Engg	Electronics	EC
Information Tech, IT, Info Tech	Information Tech	IT
Mechanical, Mech Engg	Mechanical	ME

D: Merge Mismatched Fields

Objective: Perform a vertical Union of Clean 1 (Term 1) and Clean 2 (Term 2), resolve mismatched column headers (such as Std_ID vs Student_ID, Score vs Marks_Obtained), and unify them into a single schema.

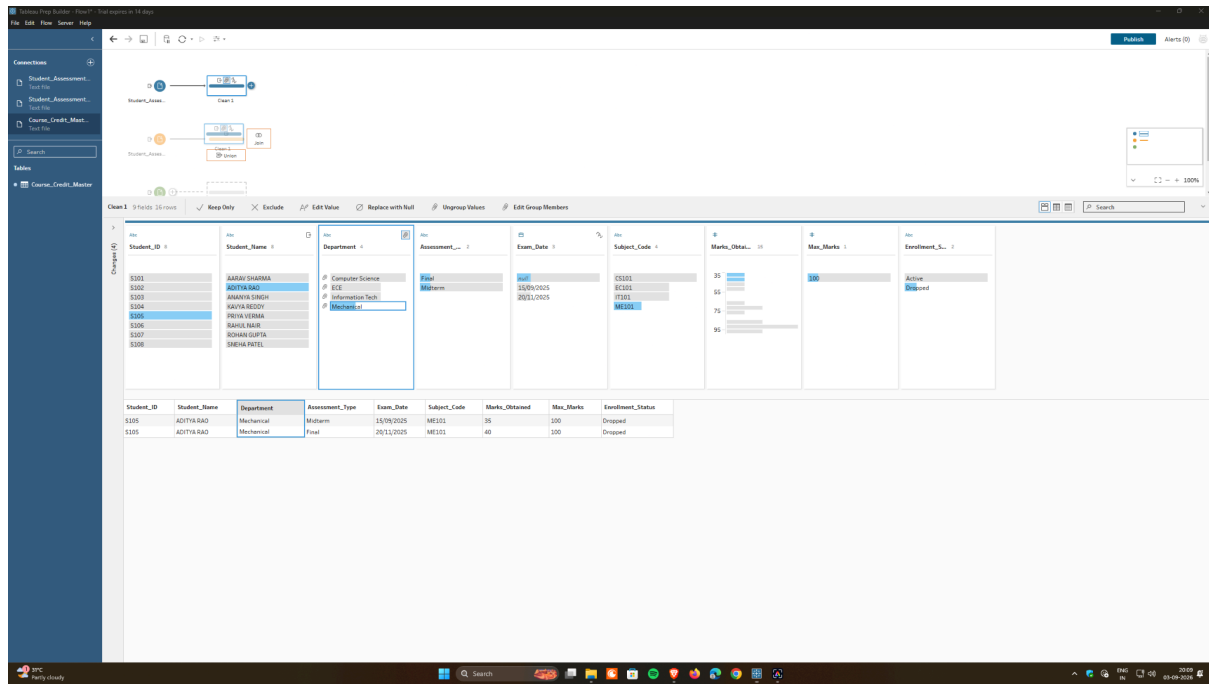
When combining datasets originating from different semesters, departments, or source systems, identical information often resides under dissimilar column names. In Tableau Prep, unmerged fields create sparse datasets with duplicate null columns. Merging mismatched fields consolidates those columns without requiring upstream source file modifications.

Step 1: Create the Union Step on the Canvas

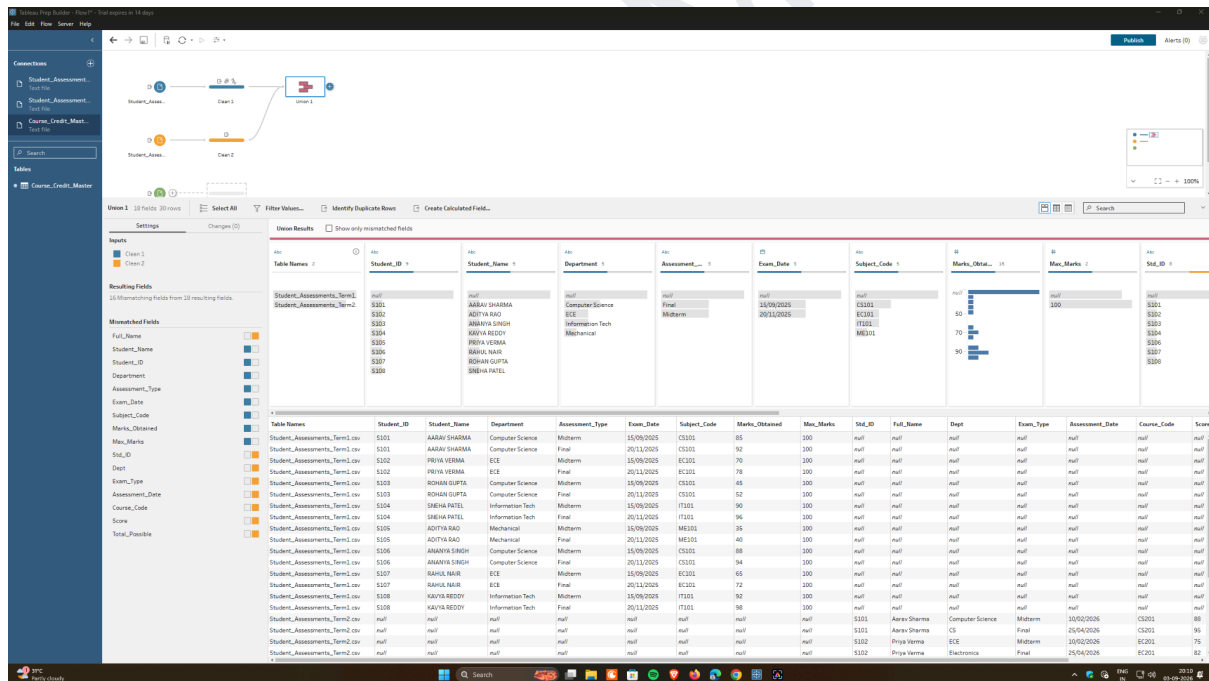
1. On the Flow canvas, click and drag the Clean 2 step icon over to the Clean 1 step Icon.

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2. When the drop zone appears, hover over Union and release the mouse. 3. A new Union 1 step is created, connecting both transformation branches.

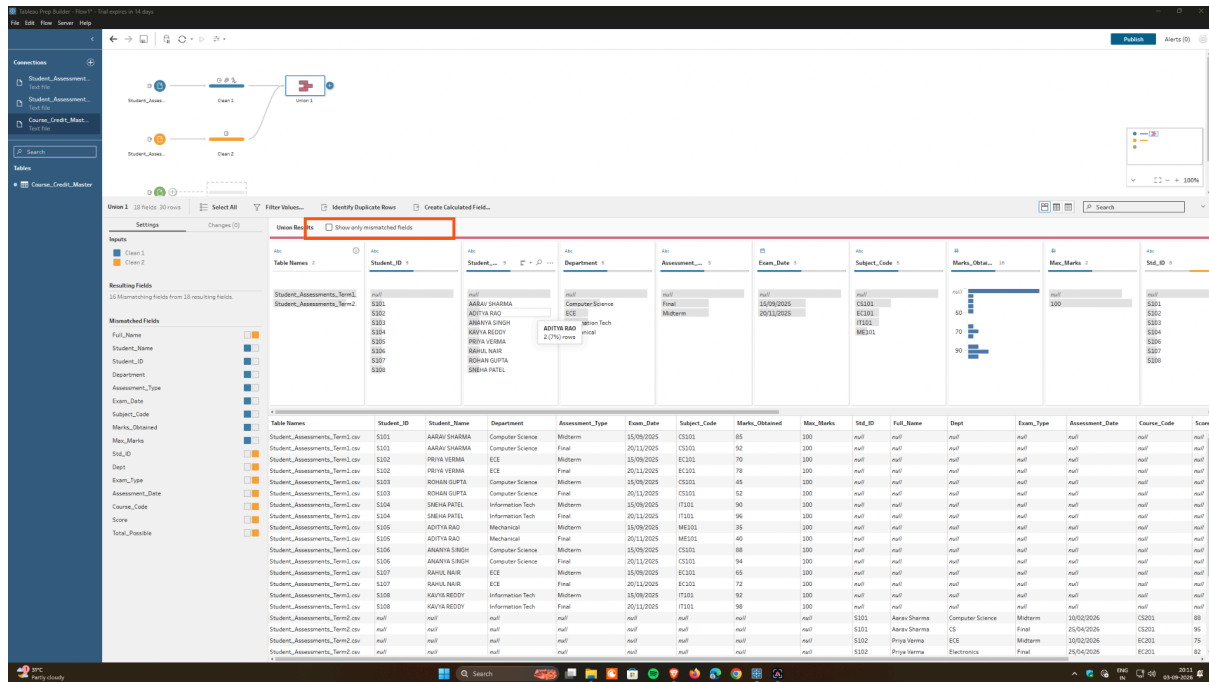


Step 2: Inspect Mismatched Fields in the Union Pane

1. Click on the Union 1 step.
2. In the bottom Union summary pane, look at the Mismatched Fields section.

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3. Notice that because the two files used different naming conventions, Tableau Prep lists non-matching pairs (indicated by yellow and blue bars), such as:

- o Student_ID (Term 1) vs Std_ID (Term 2)
- o Student_Name (Term 1) vs Full_Name (Term 2)
- o Department (Term 1) vs Dept (Term 2)
- o Assessment_Type (Term 1) vs Exam_Type (Term 2)
- o Exam_Date (Term 1) vs Assessment_Date (Term 2)
- o Subject_Code (Term 1) vs Course_Code (Term 2)
- o Marks_Obtained (Term 1) vs Score (Term 2)
- o Max_Marks (Term 1) vs Total_Possible (Term 2)

Step 3: Merge the Mismatched Fields

1. In the Mismatched Fields list (or the union profile pane):
 - o Click and drag Std_ID directly on top of Student_ID to merge them into one column.
 - o Drag Full_Name onto Student_Name.
 - o Drag Dept onto Department.

The screenshot displays the Tableau Desktop interface. On the left, the 'Data Model' pane shows a fact table 'Student_Assessments' connected to two dimension tables: 'Students' and 'Assessments'. The 'Students' table has fields like 'Student_ID', 'Full_Name', 'Department', and 'Enrollment_Status'. The 'Assessments' table has fields like 'Assessment_Type', 'Exam_Date', 'Subject_Code', 'Marks_Obtained', 'Max_Marks', 'Std_ID', 'Full_Name', 'Dept', 'Exam_Type', 'Assessment_Date', 'Course_Code', and 'Enrollment_Status'. The 'Marks_Obtained' field is highlighted in the 'Assessments' table.

The main workspace shows a dashboard with a bar chart titled 'Marks_Obtained' and a table below it. The bar chart shows 'Marks_Obtained' on the y-axis (ranging from 0 to 100) and 'Exam_Type' on the x-axis (with categories 'Final' and 'Midterm'). The table below the chart lists the data points for each exam type, showing 'Marks_Obtained' and 'Max_Marks' for each 'Std_ID' and 'Full_Name'.

The table columns are: Student_ID, Student_Name, Department, Assessment_Type, Exam_Date, Subject_Code, Marks_Obtained, Max_Marks, Std_ID, Full_Name, Dept, Exam_Type, Assessment_Date, Course_Code, and Enrollment_Status.

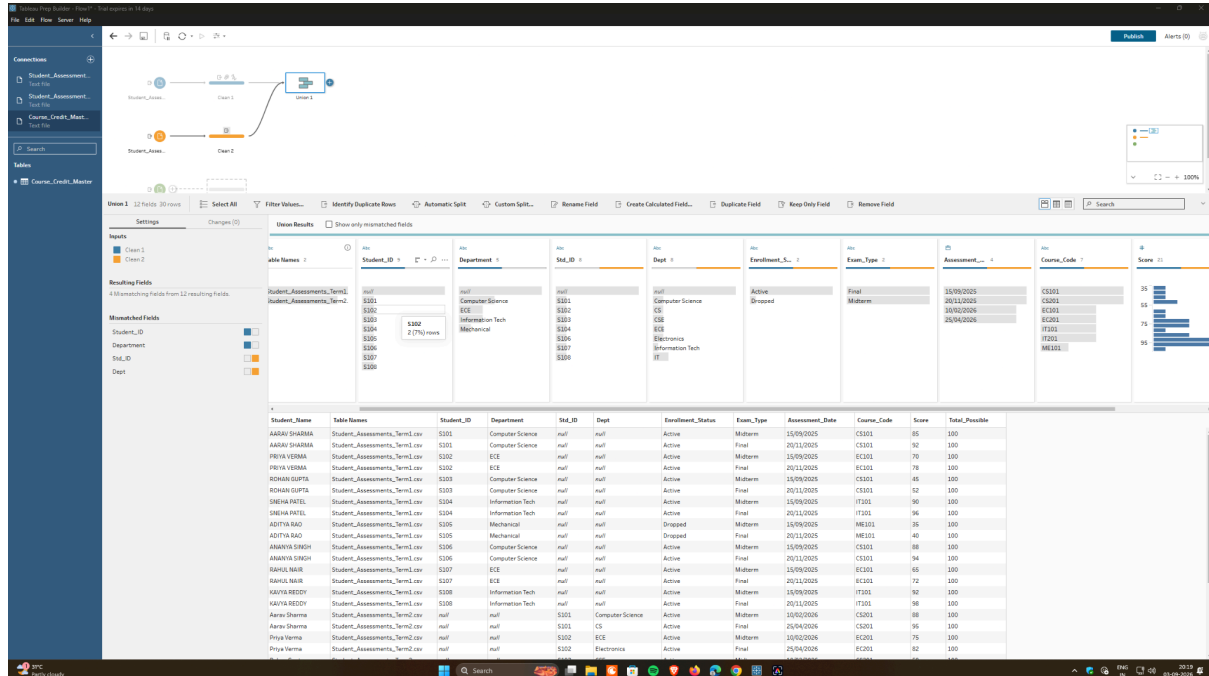
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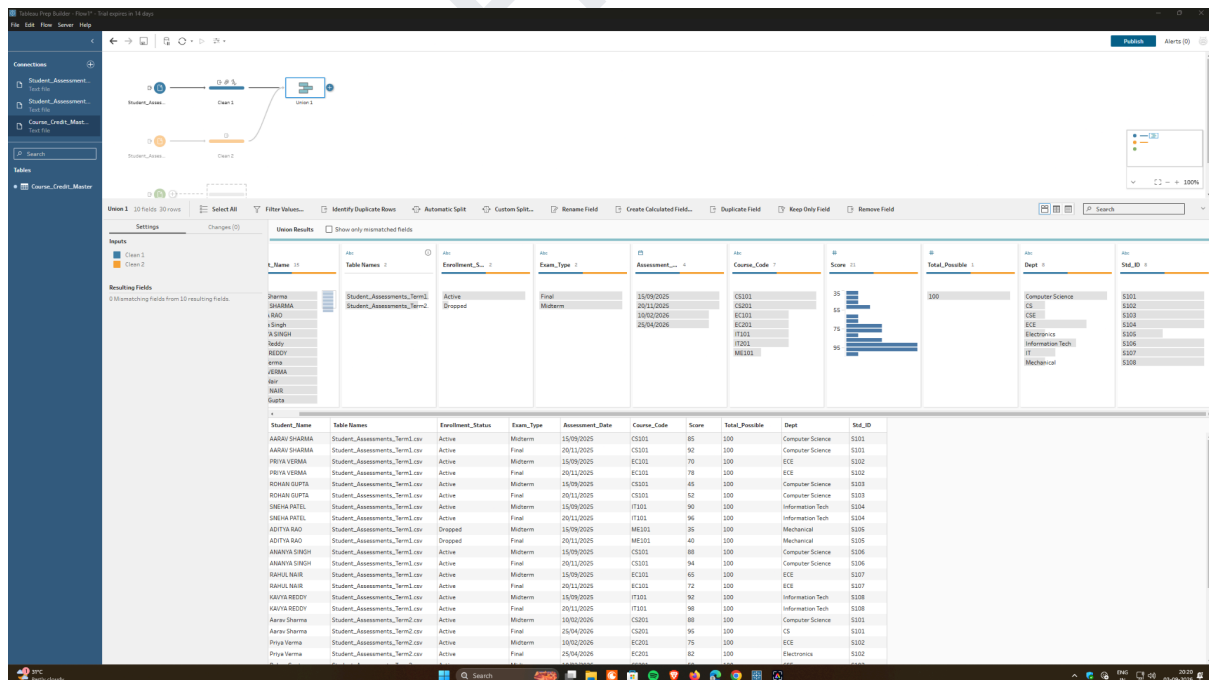
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- o Drag Assessment_Date onto Exam_Date.
- o Drag Course_Code onto Subject_Code.
- o Drag Score onto Marks_Obtained.
- o Drag Total_Possible onto Max_Marks.



2. Notice that as you merge them, the mismatched count drops to 0, and each unified column displays both source colors.



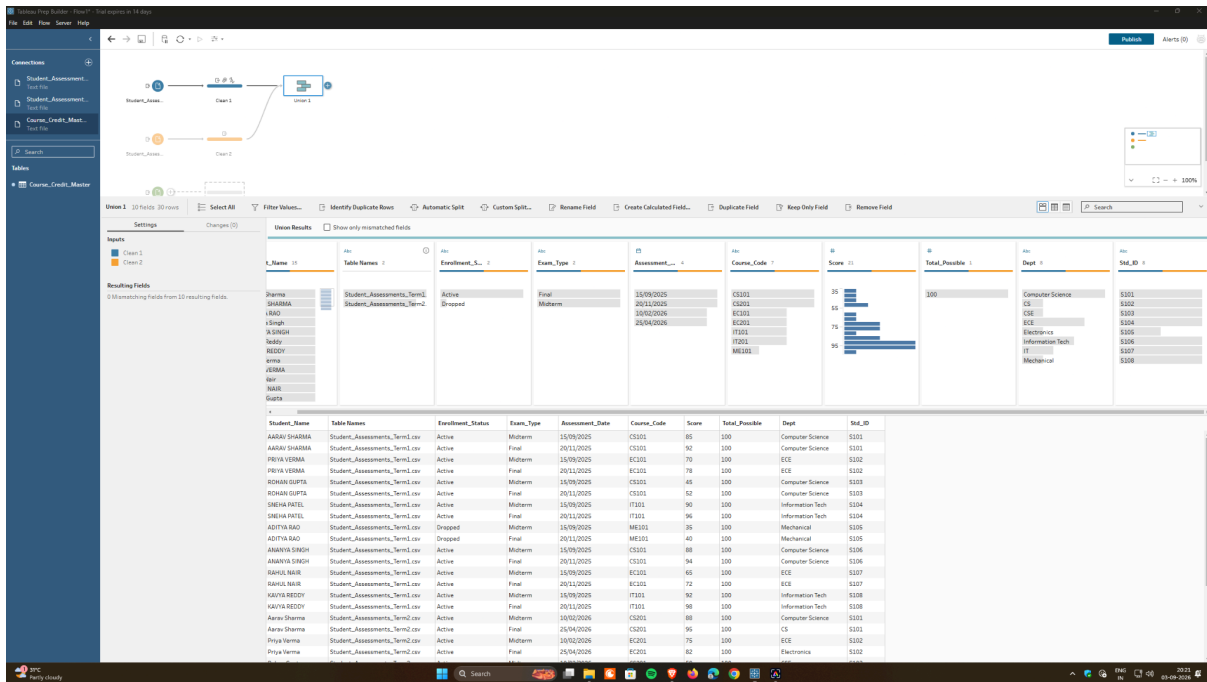
Step 4: Add a Clean Step Post-Union

1. Click the + icon next to the Union 1 node.

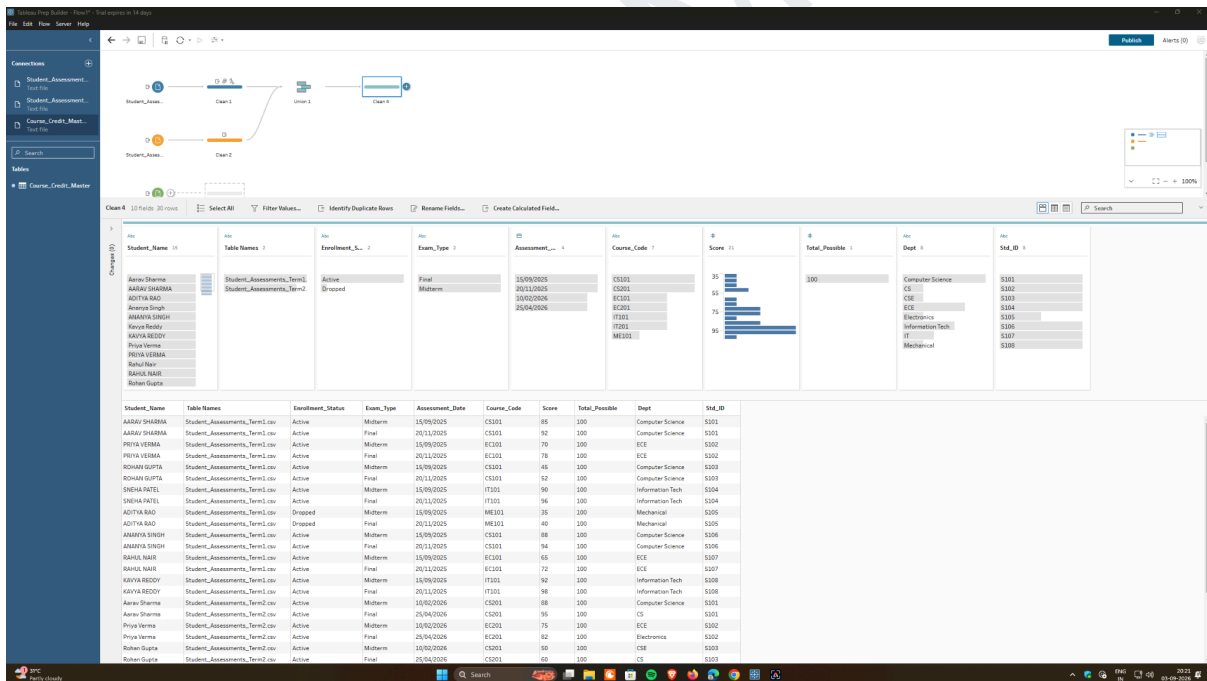
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2. Select Clean Step (creates Clean 3). In Screenshot it is showing Clean 4 because I have created clean step for Course_Credit_Master.csv too.

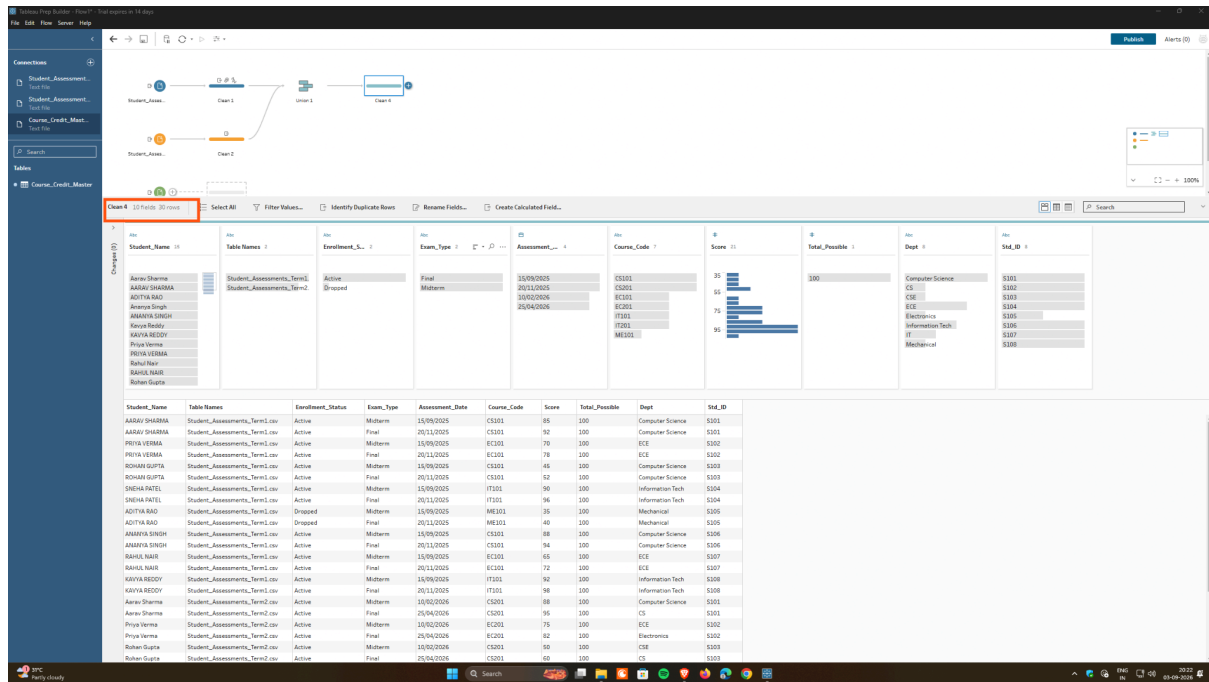


3. Verify that the combined dataset now contains 30 total rows across 10 unified columns (including Table Names).

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E: Perform Aggregations in Prep

Objective: Create an Aggregate Step in Tableau Prep to summarize granular, exam-level test scores into student-level cumulative metrics (Total Marks, Average Score, and Total Assessment Count) per term

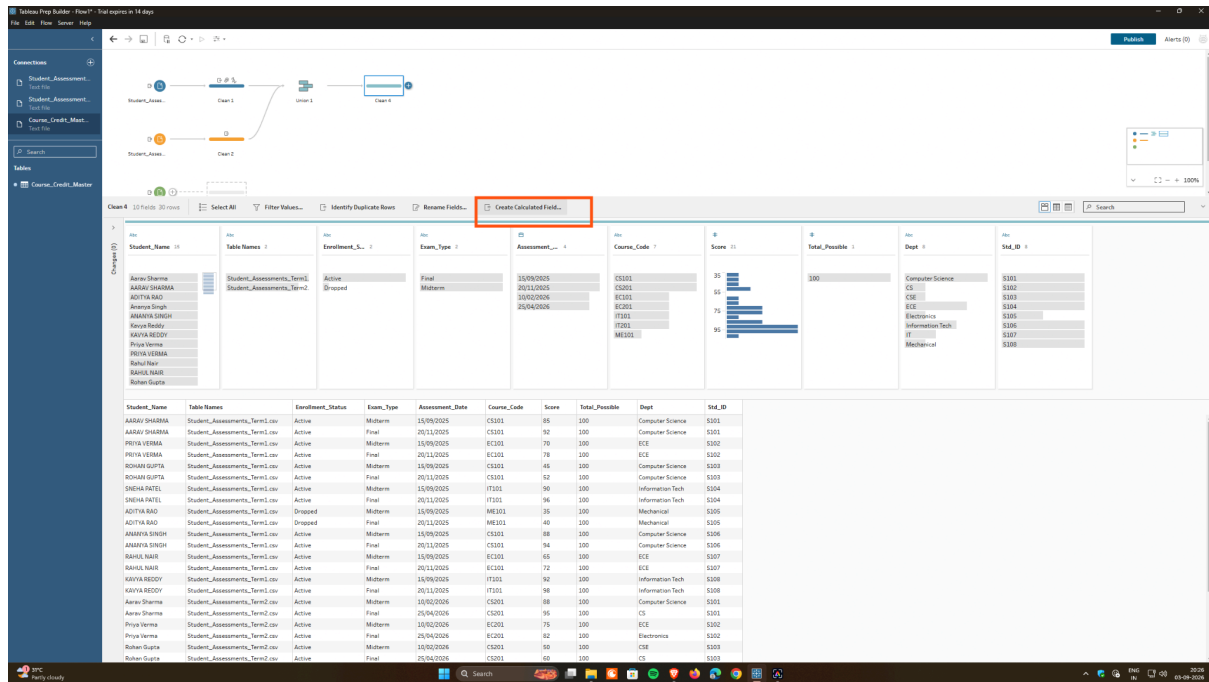
Raw assessment tables log every individual examination (Midterm, Final) on a separate row. By performing an aggregation step directly in Tableau Prep, we roll up individual row events into pre-computed summary statistics, reducing data redundancy and generating student scorecard baselines prior to dashboarding.

Step 1: Add a Term Identifier Field in Clean 3

1. Click on the Clean 3 step (the clean step created after Union 1).
2. In the toolbar above the Profile cards, click Create Calculated Field You have to click three dots on the toolbar.

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3. Set the Field Name to: Term

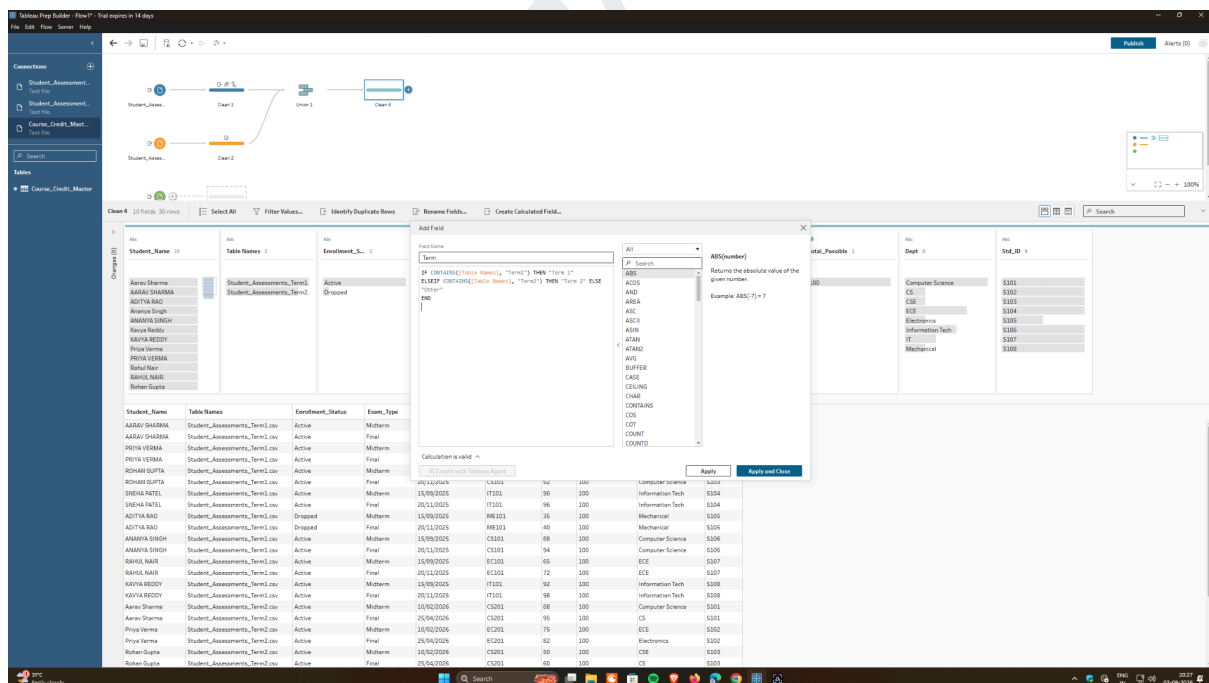
4. Enter the formula:

Code snippet

IF CONTAINS([Table Names], "Term1") THEN "Term 1"

ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2" ELSE "Other"

END



5. Click Save.

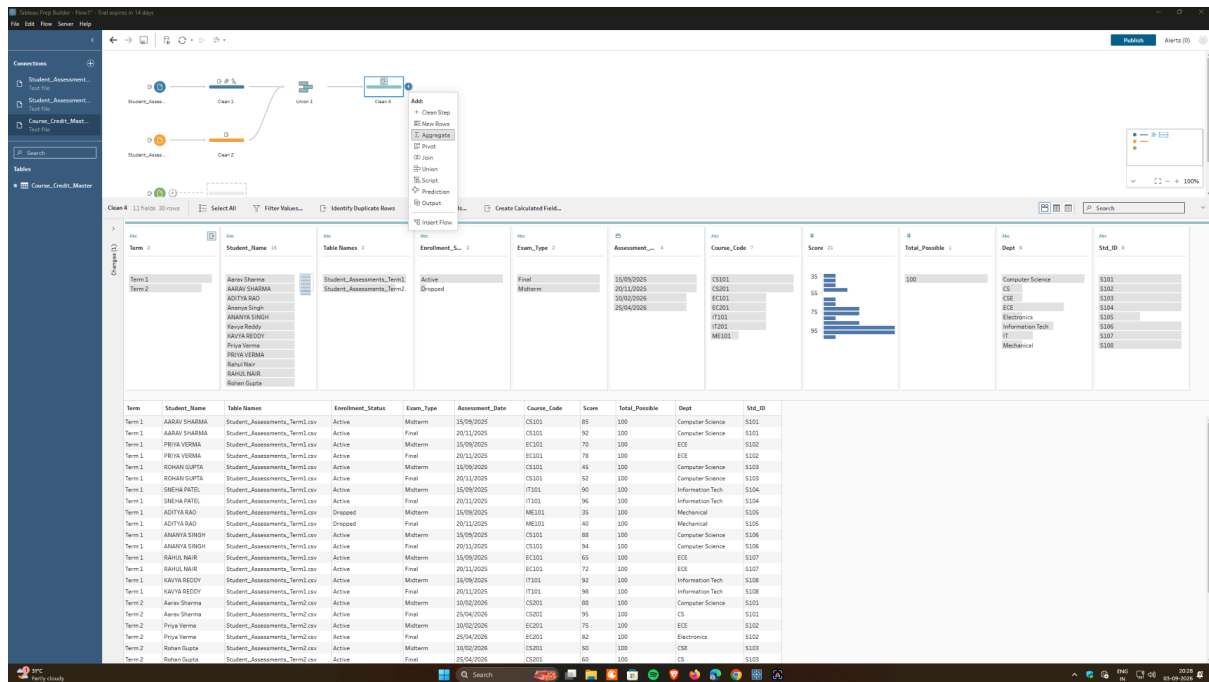
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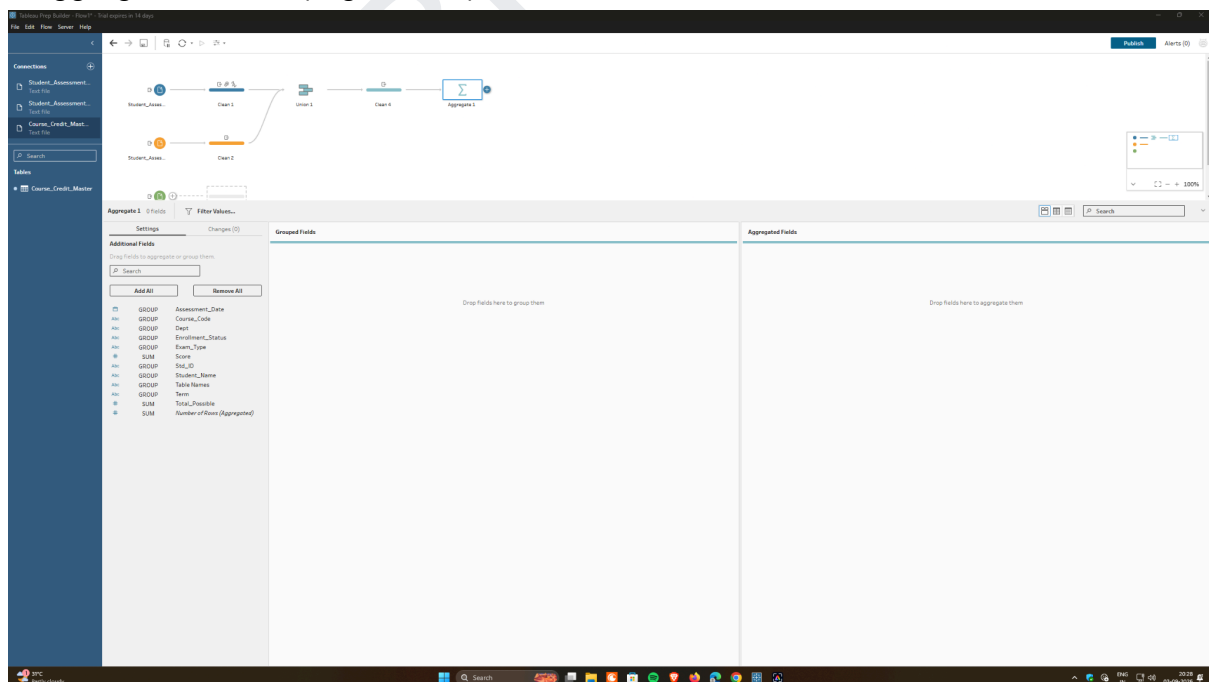
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Step 2: Add an Aggregate Step to the Flow Canvas

1. Hover over the Clean 3 node on the Flow canvas.
2. Click the + icon next to it.
3. Select Aggregate (represented by the summation icon Σ)



4. An Aggregate 1 step will appear, showing two drop zones in the configuration pane:
 - o Grouped Fields (Left Zone)
 - o Aggregated Fields (Right Zone)



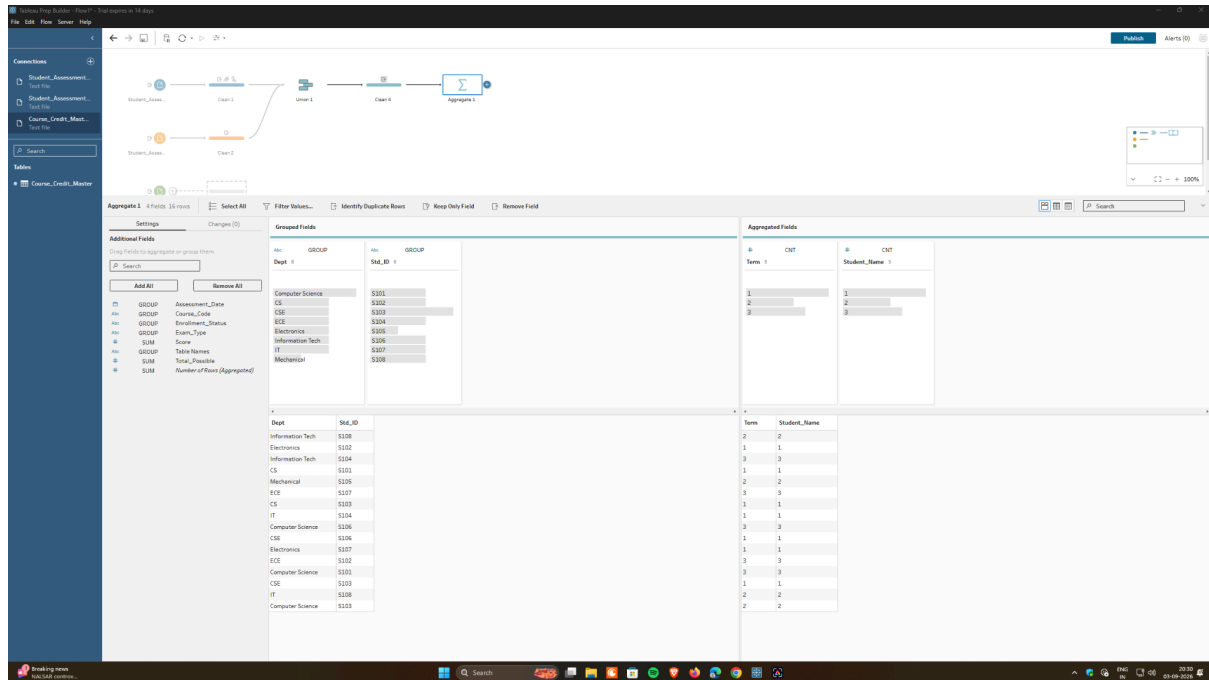
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Step 3: Configure Grouped Fields (Dimensions)

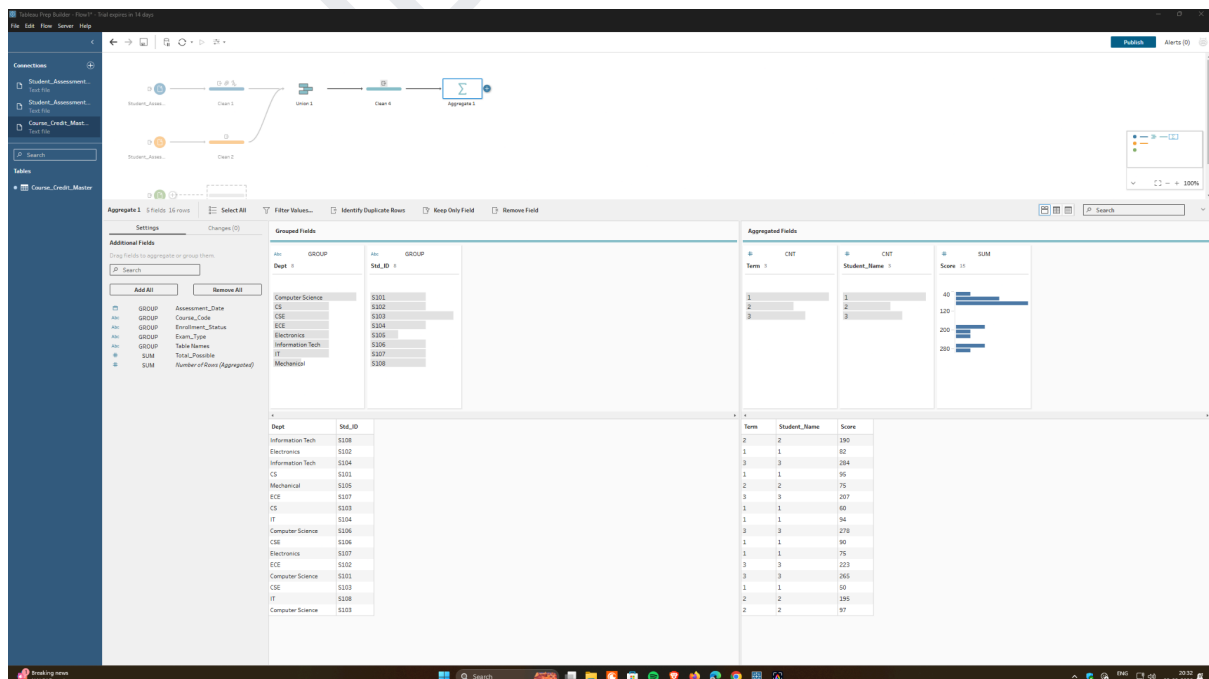
From the left-hand column list in the Aggregate step pane Your Column names could be different based on how you did previous steps of Union:

1. Drag Student_ID into the Grouped Fields box.
2. Drag Student_Name into the Grouped Fields box.
3. Drag Department into the Grouped Fields box.
4. Drag Term into the Grouped Fields box.



Step 4: Configure Aggregated Fields (Measures & Metrics)

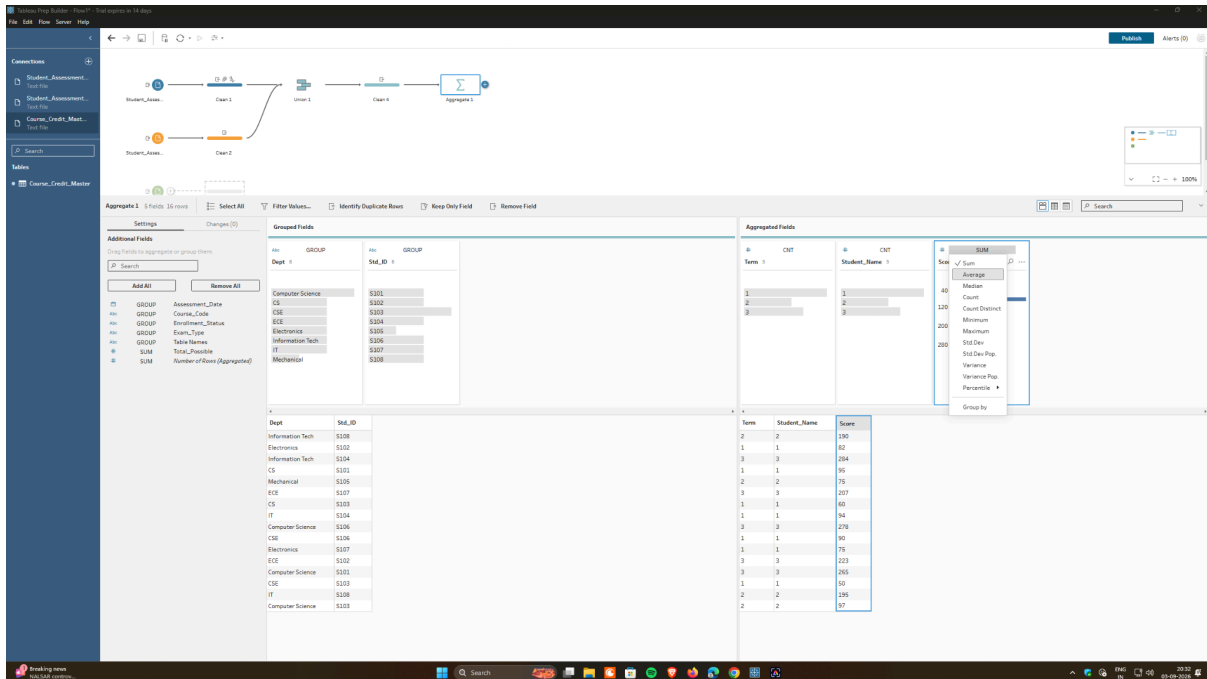
1. Total Marks: Drag Marks_Obtained into the Aggregated Fields box. (Ensure aggregation operation is set to Sum).



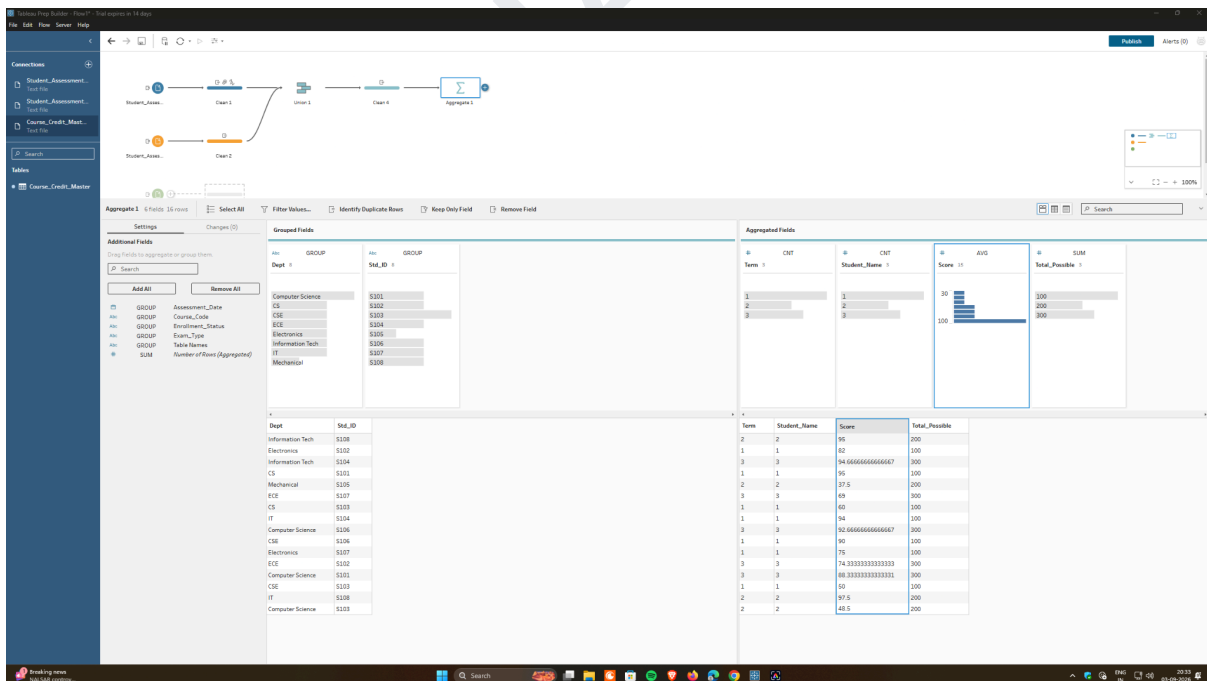
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2. Average Score: Drag Marks_Obtained into the Aggregated Fields box a second time --> click on Sum --> change it to Average. There will be no field for dragging by any chance just change SUM to AVERAGE of the existing one.



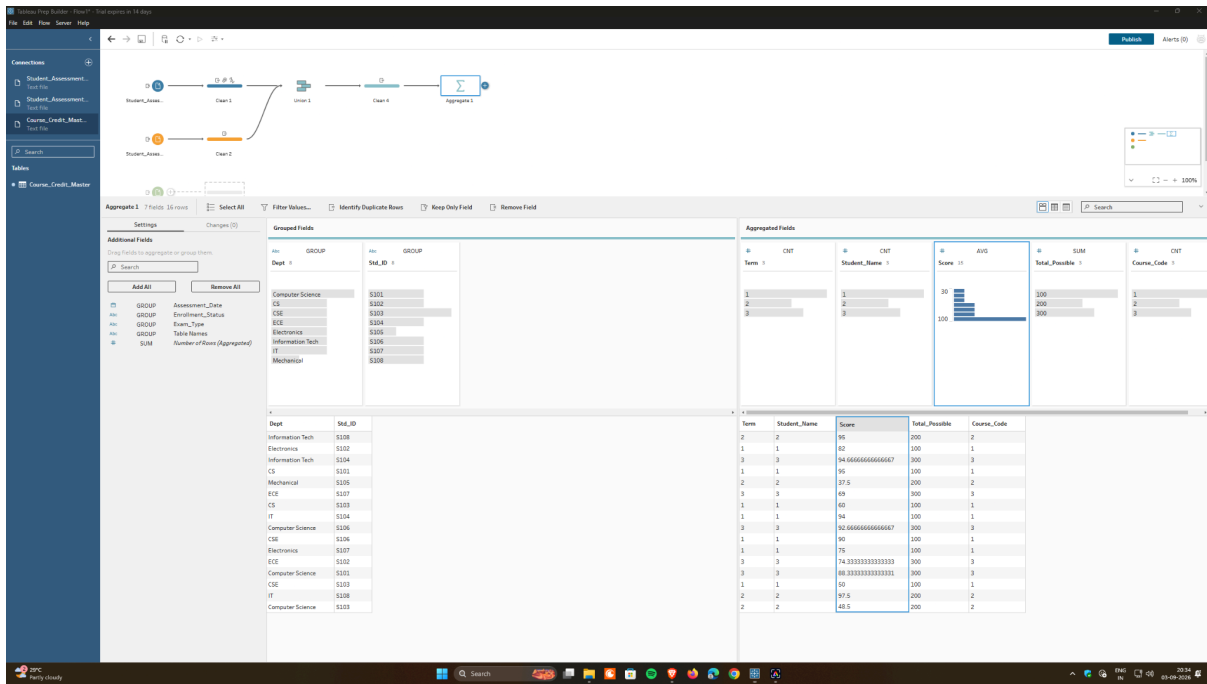
3. Total Possible Marks: Drag Max_Marks into the Aggregated Fields box --> keep as Sum.



4. Assessment Count: Drag Assessment_Type (or Subject_Code) into the Aggregated Fields box --> set aggregation to Count (or Count Distinct).

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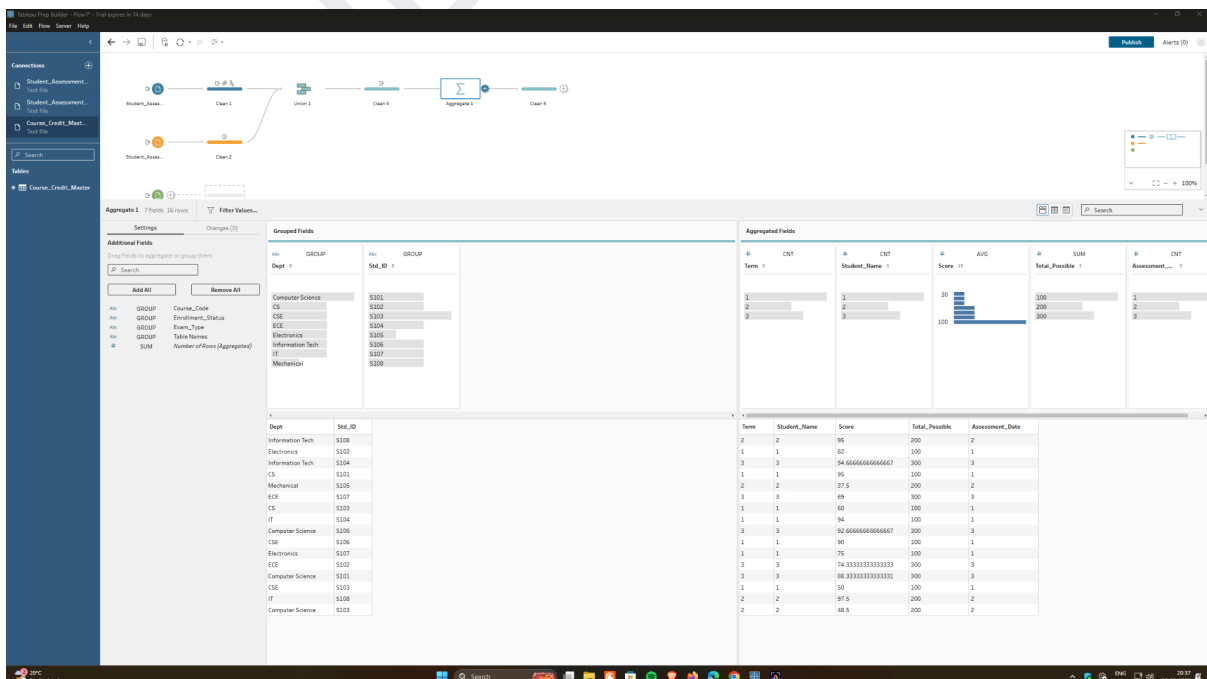
Step 5: Rename Aggregated Output Fields

In the Aggregated Fields panel, click on each field name to rename them cleanly:

- Marks_Obtained (Sum) --> Total_Marks_Obtained
- Marks_Obtained (Average) --> Average_Score
- Max_Marks (Sum) --> Total_Max_Marks
- Assessment_Type (Count) --> Assessments_Taken

Step 6: Add a Clean Step to Inspect Aggregated Output

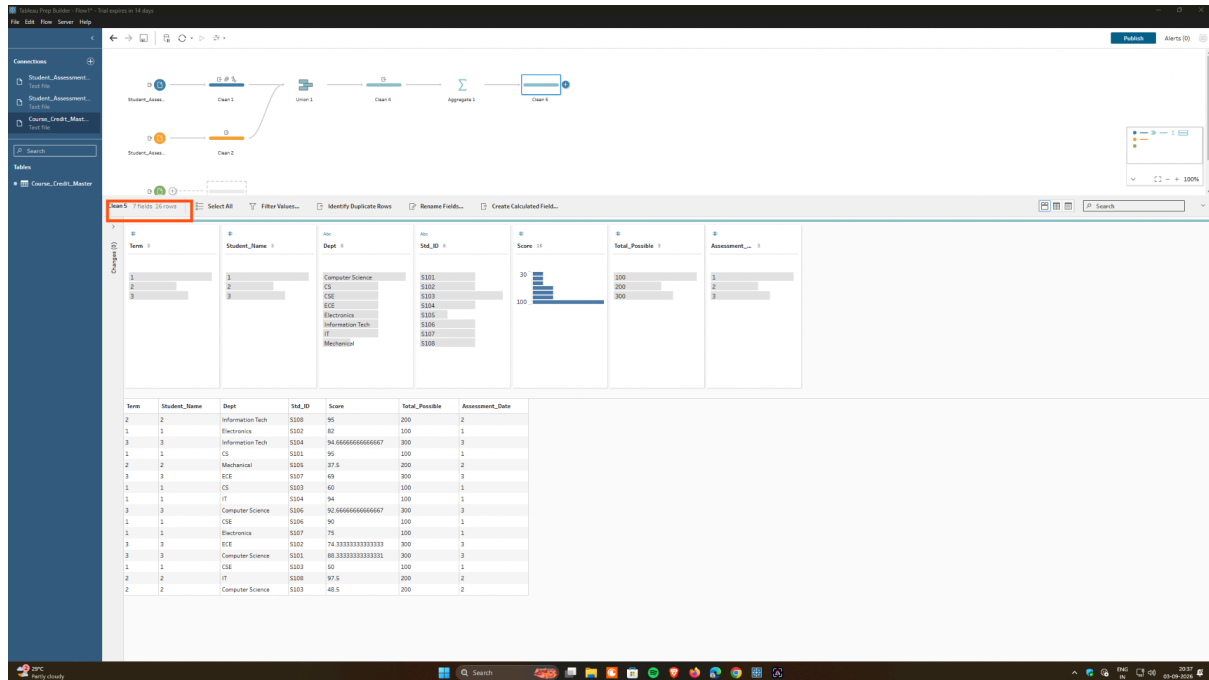
1. Click the + icon next to the Aggregate 1 node --> select Clean Step (creates Clean 4).



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2. Look at the Profile Pane: o The row count collapses from 30 individual assessment rows into 15 summary records or more (one row per student per term).



F: Apply Filters and Transformations

Objective: Remove inactive/dropped records, compute weighted percentages, assign automated grading tiers, and derive academic standing indicators using calculated fields in Tableau Prep.

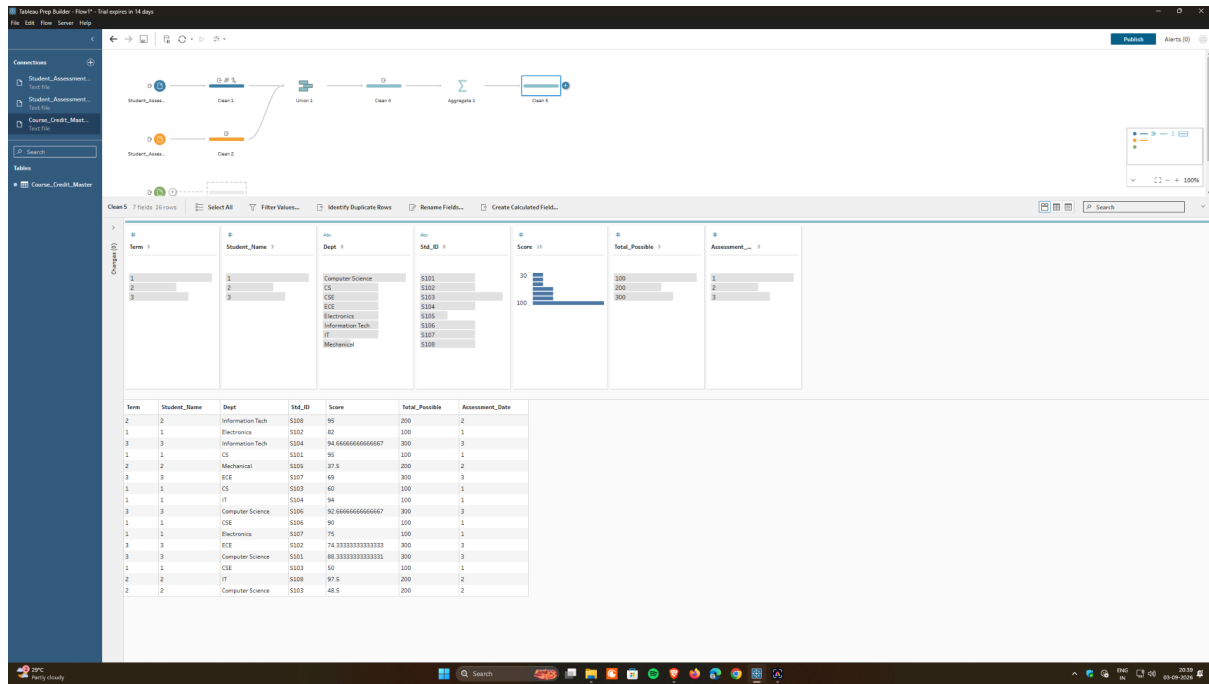
Not all raw records should enter downstream analytical dashboards (e.g., students who dropped out mid-term skew department averages). Transforming raw totals into percentage metrics and standardized letter grades directly inside the data prep pipeline ensures that business logic is applied once and remains consistent across all downstream reporting tools.

Step 1: Open the Transformation Step (Clean 4)

1. Click on the Clean 4 step connected after your Aggregate 1 step on the Flow canvas.
2. The Profile Pane below will show your aggregated columns: Student_ID, Student_Name, Department, Term, Total_Marks_Obtained, Average_Score, Total_Max_Marks, and Assessments_Taken. (Naming Will be different and completely depends on how you performed the Union Step)

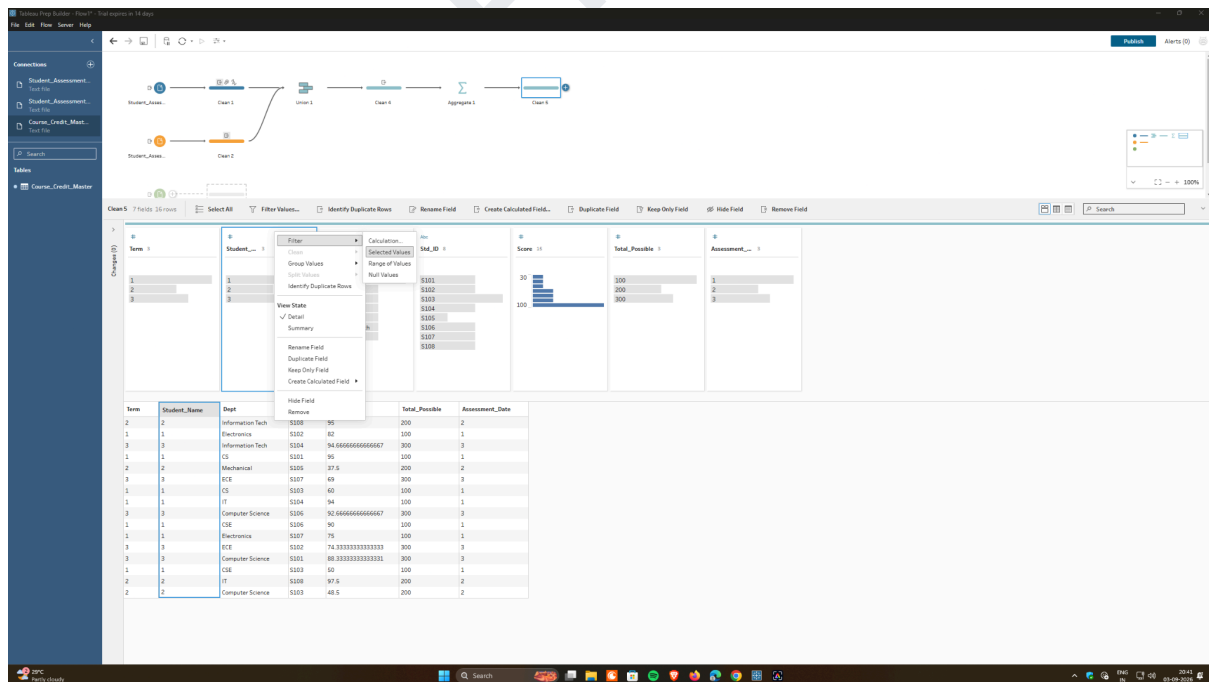
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Step 2: Filter Out Inactive / Dropped Student Records

1. If Enrollment_Status is present on a profile card:
 - o Click on the card --> click the value Dropped --> right-click and select Exclude (or click the filter icon at the top of the card and check only Active).
2. (Alternative via Filter calculation):
 - o In the top toolbar, click Filter Values --> Selected Values....

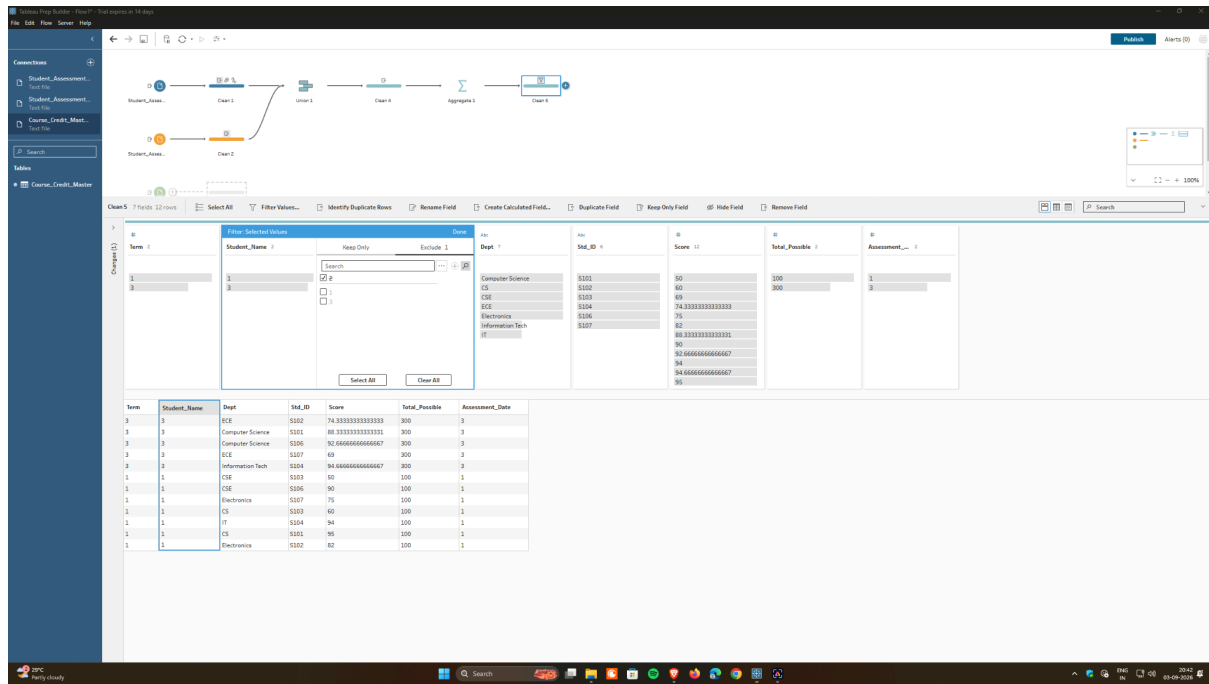


- Exclude Dropped so only Active students remain in the active pipeline.

SHUBHAM KARAPE
T085

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Step 3: Create the Normalized Term Percentage Metric

1. In the toolbar above the Profile cards, click Create Calculated Field.

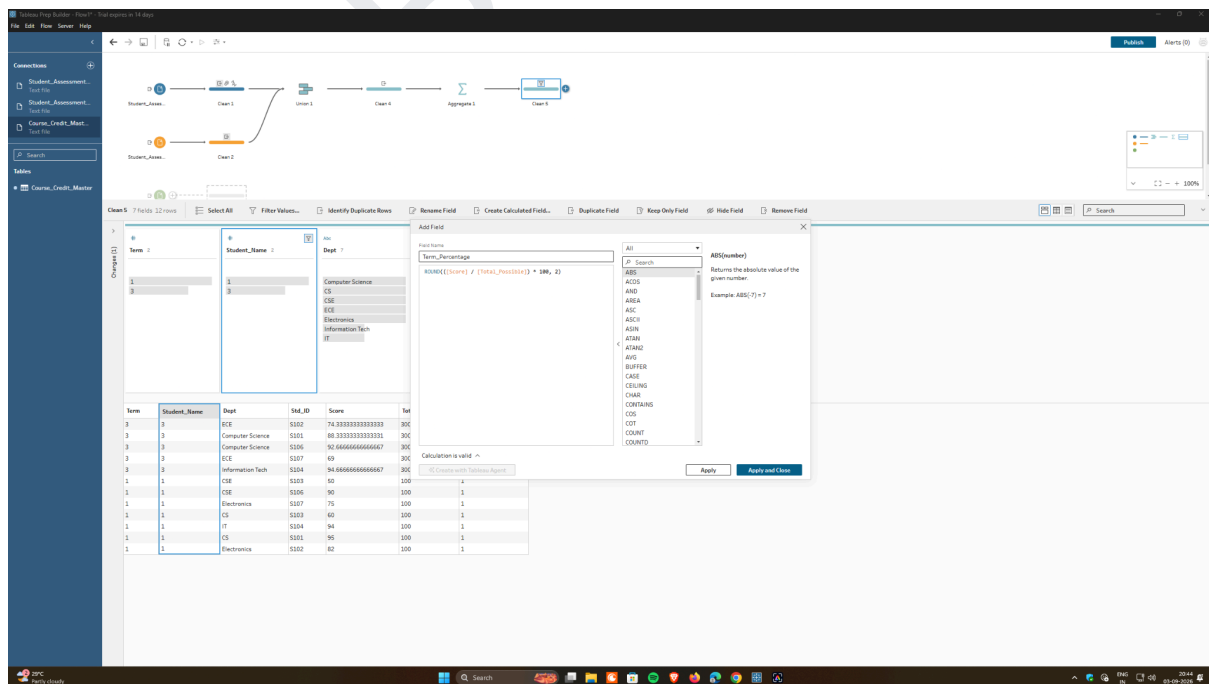
2. Field Name: Term_Percentage

3. Tableau Prep Formula:

Code snippet

$$\text{ROUND}(\left(\frac{[\text{Total_Marks_Obtained}]}{[\text{Total_Max_Marks}]} \times 100, 2\right) \text{ ROUND}(\left(\frac{[\text{Score}]}{[\text{Total_Possible}]} \times 100, 2\right) \{ \text{This formula can be used if you couldn't able to rename your aggregate fields.} \}$$

4. Click Save.



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Step 4: Create the Categorical Letter Grade Transformation

1. In the toolbar, click Create Calculated Field.

2. Field Name: Letter_Grade

3. Tableau Prep Formula:

Code snippet

```
IF [Term_Percentage] >= 85 THEN "A (Distinction)"
```

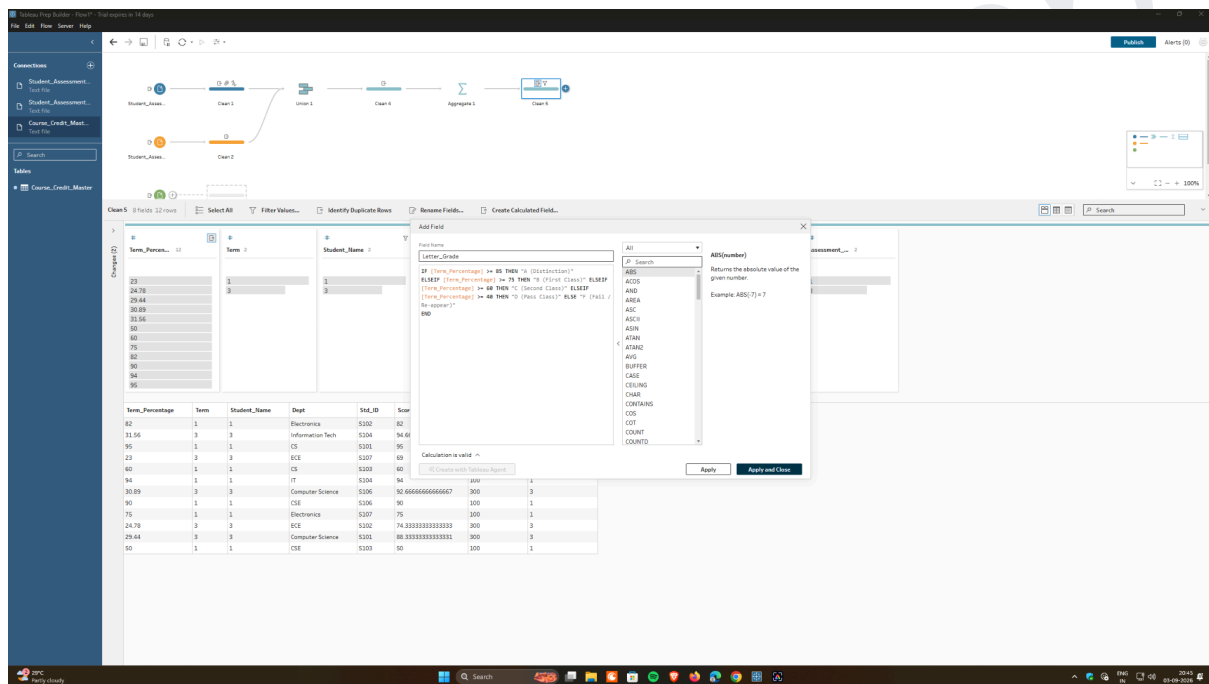
```
ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)" ELSEIF
```

```
[Term_Percentage] >= 60 THEN "C (Second Class)" ELSEIF [Term_Percentage] >=
```

```
40 THEN "D (Pass Class)" ELSE "F (Fail / Re-appear)"
```

END

4. Click Save.



Step 5: Create Academic Standing Compliance Flag

1. Click Create Calculated Field.

2. Field Name: Academic_Standing

3. Tableau Prep Formula:

Code snippet

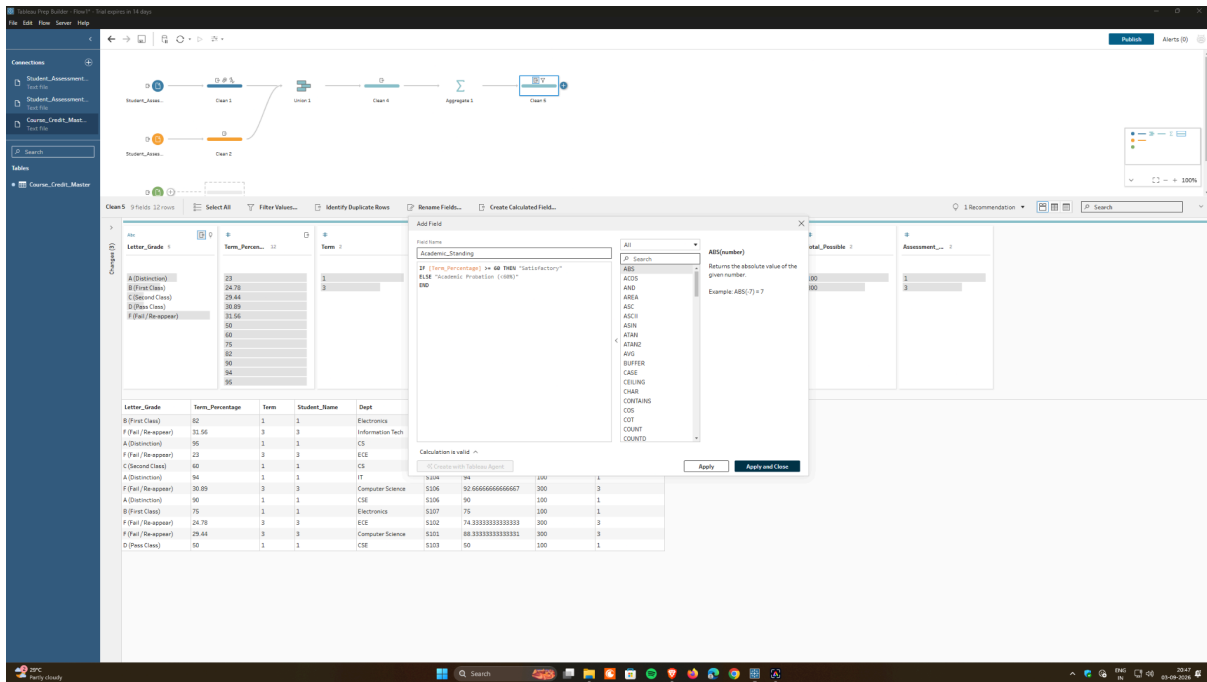
```
IF [Term_Percentage] >= 60 THEN "Satisfactory"
```

```
ELSE "Academic Probation (<60%)"
```

END

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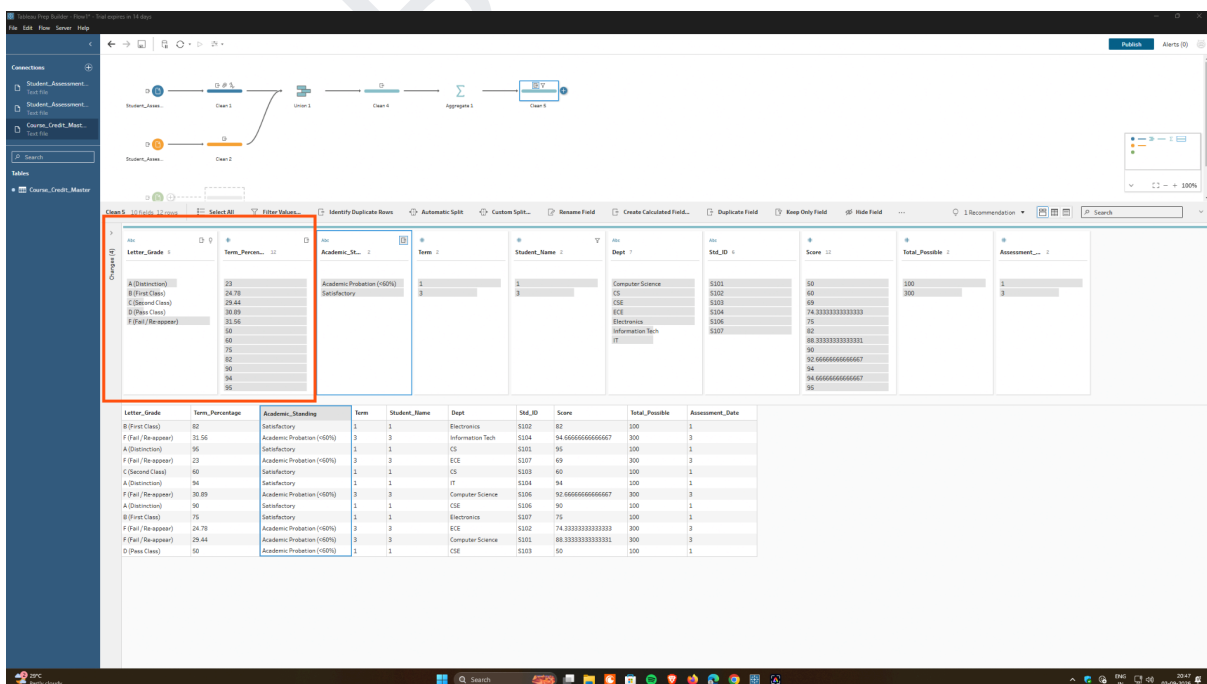


4. Click Save.

Step 6: Verify Transformed Profile Distributions

1. Look at the newly generated profile cards in Clean 4:

- o Term_Percentage: Displays a smooth numeric distribution between 0% and 100%.
- o Letter_Grade: Displays discrete categorical buckets (A (Distinction), B (First Class), etc.).
- o Academic_Standing: Highlights students requiring academic counseling.



Transformation Summary Table

Transformation	Type	Formula / Rule	Output Meaning
Filter Status	Row Exclusion	[Enrollment_Status] != 'Dropped'	Removes dropped students
Term_Percentage	Numeric Calculation	([Total_Marks_Obtained] / [Total_Max_Marks]) * 100	Normalized 0–100 scale
Letter_Grade	Conditional Dimension	IF [Term_Percentage] >= 85 THEN "A (Distinction)" ...	Standardized letter grade
Academic_Standi ng	Conditional Flag	IF [Term_Percentage] >= 60 THEN "Satisfactory" ...	Compliance warning tag

G: Create and Automate Data Preparation Flows

Objective: Configure an Output Step in Tableau Prep Builder to publish the fully transformed pipeline to an extract file (.hyper or .csv), and learn how to automate or execute the flow on demand.

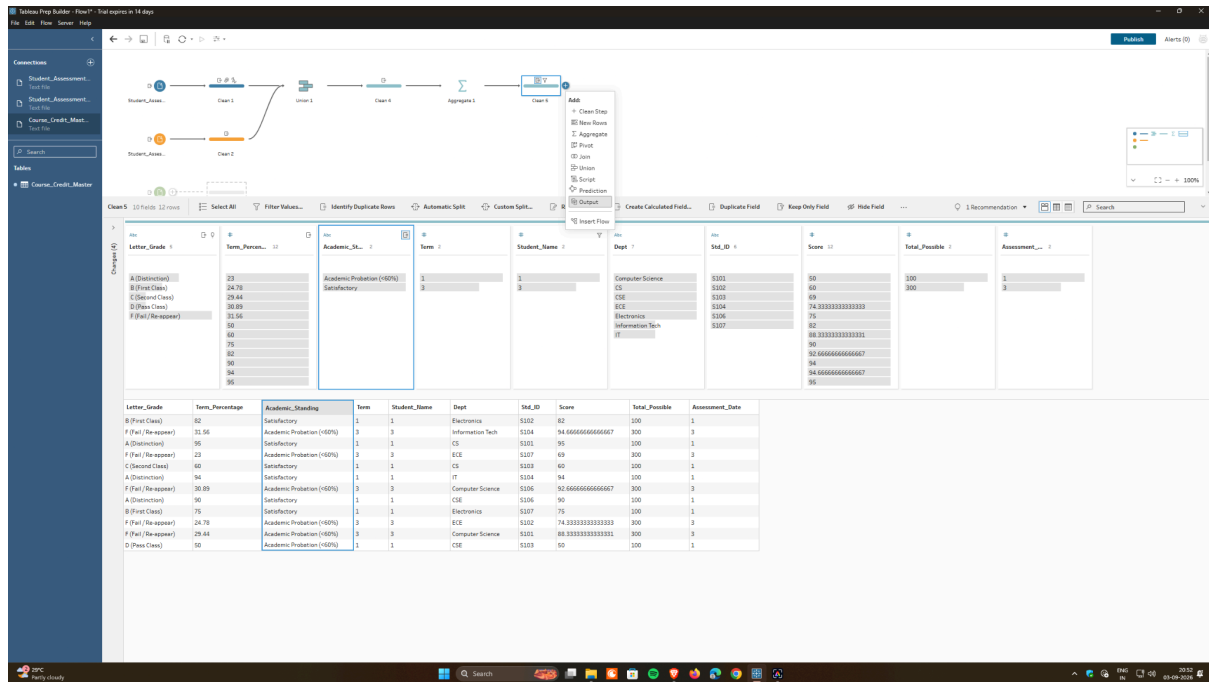
Data preparation workflows must be repeatable and production-ready. Configuring a dedicated output node ensures that whenever new assessment scores arrive, running the flow automatically executes all trimming, grouping, merging, filtering, and scoring calculations in seconds without any manual rework.

Step 1: Add an Output Step to the End of the Pipeline

1. On the Flow canvas, locate the final cleaning step (Clean 4).
2. Hover over the Clean 4 icon and click the + icon.
3. Select Output (represented by the download/save arrow icon).
4. An Output step node will appear connected directly to Clean 4.

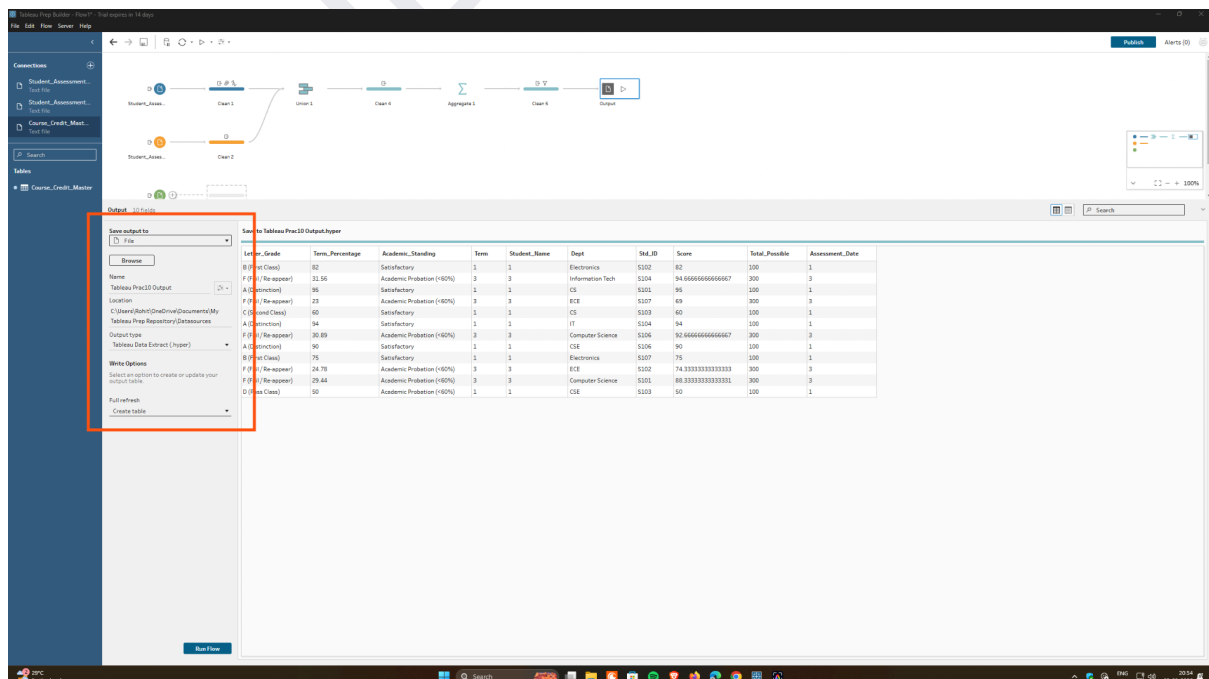
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Step 2: Configure Output Settings in the Bottom Pane

1. Click on the Output node to open its configuration panel below:
 - o Save output to: Select File (or Tableau Server / Tableau Cloud).
 - o Output type: Select Tableau Data Extract (.hyper) for high-speed dashboarding, or Comma Delimited Text (.csv) for universal export.
 - o Location: Click Browse and choose the directory where you want the final dataset saved.
 - o File Name: Enter Student_Performance_Master_Cleaned.
 - o Write Option: Select Create table (or Append to table if updating existing records).



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Step 3: Run the Data Prep Flow

1. In the Output pane at the bottom-left, click the blue Run Flow button (or click the play icon $\blacktriangleright$ directly on the Output node on the canvas).
2. Tableau Prep will process all upstream nodes (Cleaning $\rightarrow$ Grouping $\rightarrow$ Unioning $\rightarrow$ Merging $\rightarrow$ Aggregating $\rightarrow$ Scoring).
3. A success notification will appear stating:

"The flow ran successfully in X seconds."

The screenshot shows the Tableau Prep Builder interface. The canvas displays a flow with nodes: Student_Assess, Clean 1, Union 1, Clean 2, Aggregates 1, Clean 3, and Output. The Output pane at the bottom left shows the 'Run Flow' button. A dialog box titled 'Finished Running Flow' is centered on the screen, indicating the flow completed successfully. In the background, a data table is visible with columns: Letter_Grade, Term_Percentage, Academic_Standing, Term, Student_Name, Dept, Std_ID, Score, Total_Possible, and Assessment_Date. The table contains 10 rows of data.

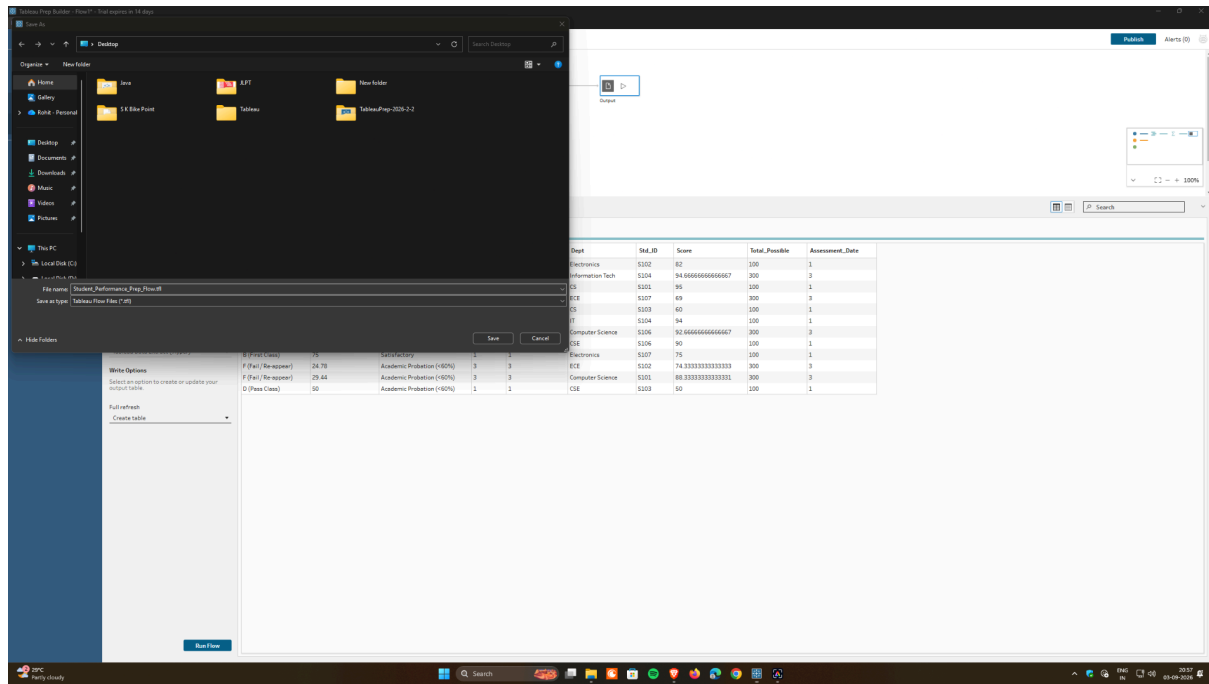
Letter_Grade	Term_Percentage	Academic_Standing	Term	Student_Name	Dept	Std_ID	Score	Total_Possible	Assessment_Date
B (First Class)	82	Satisfactory	1	1	Electronics	51502	82	100	1
F (Fail / Re appear)	31.56	Academic Probation (100%)	1	3	Information Tech	51504	34.6666666666667	100	1
A (Distinction)	95	Satisfactory	1	1				100	1
F (Fail / Re appear)	23	Academic Probation (100%)	1	3				100	1
C (Second Class)	60	Satisfactory	1	1				100	1
A (Distinction)	94	Satisfactory	1	1				100	1
F (Fail / Re appear)	30.89	Academic Probation (100%)	1	3				100	1
A (Distinction)	96	Satisfactory	1	1				100	1
B (First Class)	76	Satisfactory	1	1				100	1
F (Fail / Re appear)	24.78	Academic Probation (100%)	1	3				100	1
F (Fail / Re appear)	29.84	Academic Probation (100%)	1	3				100	1
D (Pass Class)	60	Academic Probation (100%)	1	1				100	1

Step 4: Save and Automate the Flow File

1. Go to the top menu bar $\rightarrow$ select File $\rightarrow$ Save As....
2. Name your workflow:
Plaintext
Student_Performance_Prep_Flow.tfl

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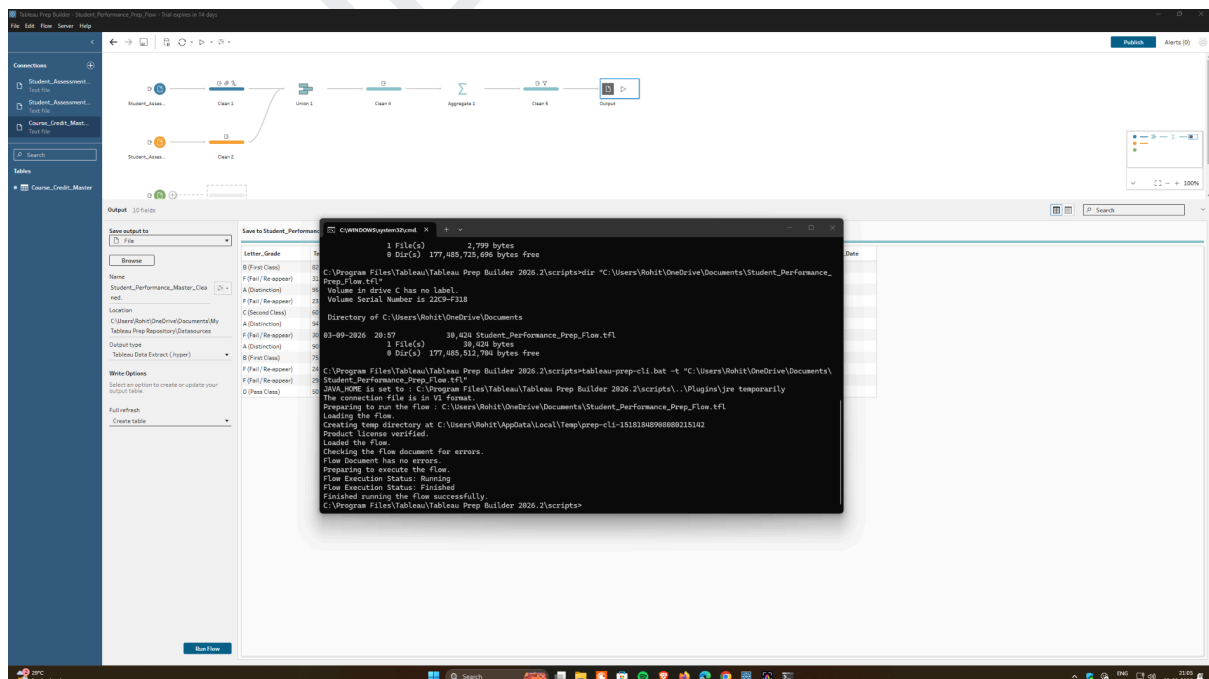
3. (Automation Options in Production):

o Command Line Execution: You can automate this flow on Windows/Mac using the Tableau Prep Command Line Interface (CLI):

Bash

```
tableau-prep-cli -t Student_Performance_Prep_Flow.tfl
```

o Tableau Prep Conductor: If connected to Tableau Cloud/Server, you can publish the flow and set a scheduled refresh (e.g., every Monday at 6:00 AM) to automatically run whenever new semester files are added.

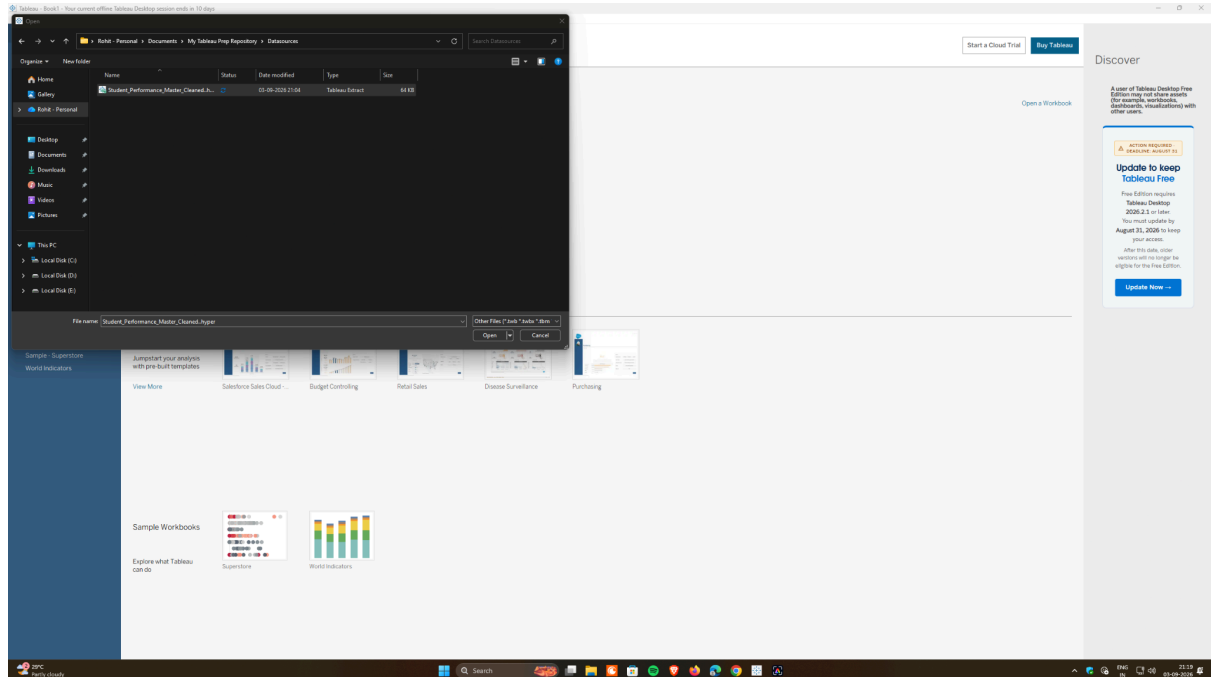


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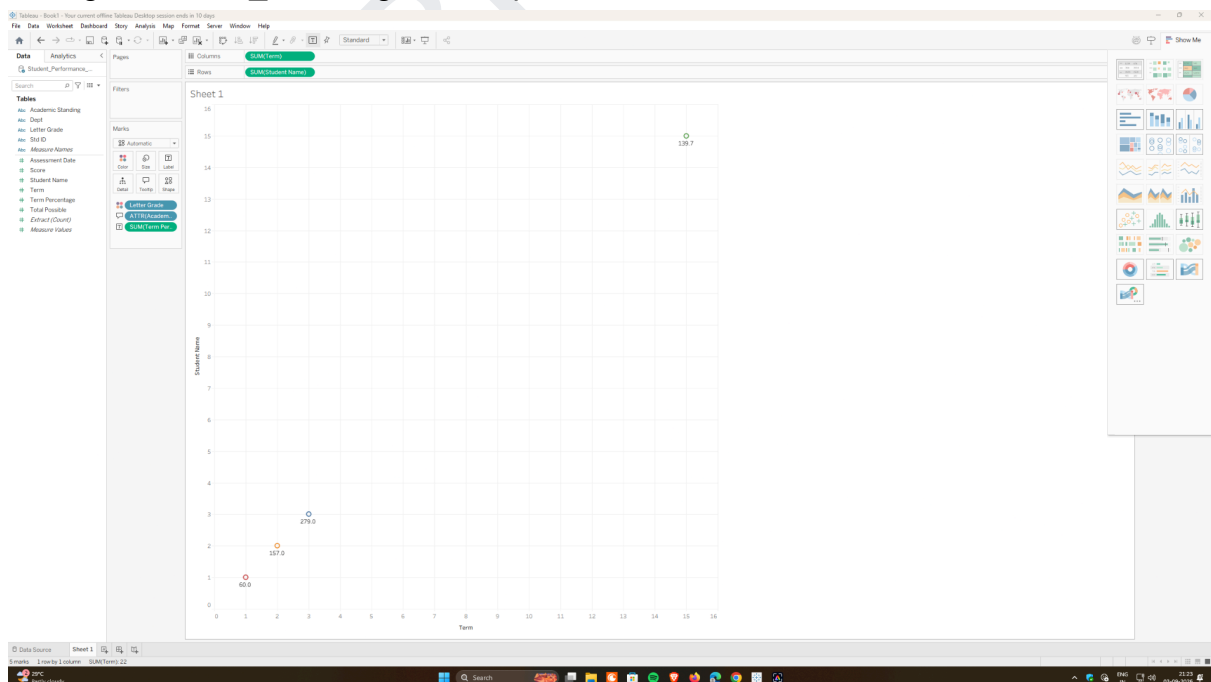
Step 5: Verify the Final Master Dataset in Tableau Desktop

1. Open Tableau Desktop.
2. Connect to the exported Student_Performance_Master_Cleaned.csv (or .hyper).

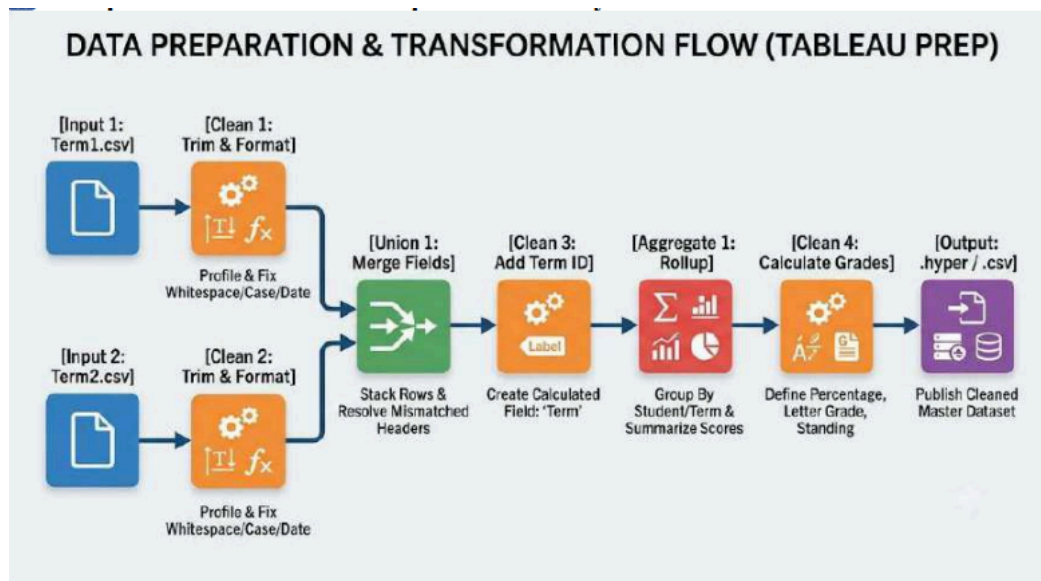


3. Build the self-performance dashboard using the pre-calculated metrics:

- o Drag Student_Name to Rows.
- o Drag Term to Columns.
- o Drag Term_Percentage to Text.
- o Drag Letter_Grade to Color.
- o Drag Academic_Standing to Tooltip.



Complete End-to-End Flow Pipeline Summary



Pipeline Stage	Step Type	Key Actions Performed	Output Result
Input Layer	Data Ingestion	Connect to Term1.csv, Term2.csv, and Course_Master.csv	Raw tables loaded on canvas
Cleaning Layer	Clean Steps 1 & 2	Trim whitespace, standardize dates, uppercase names	Structured single-source tables
Standardization	Group Values	Consolidate aliases (CS, CSE, Comp Sci → Computer Science)	Clean department dimensions
Integration Layer	Union Step	Stack Term 1 & 2; drag-and-drop mismatched headers	Unified schema (30 assessment rows)
Aggregation Layer	Aggregate Step	Group by student & term; compute Sum(Marks) and Avg(Score)	14 summary records
Business Logic	Clean Step 4	Filter dropped students; calculate %, Letter_Grade, and Standing	Analytical scorecard metrics
Export Layer	Output Step	Export to Student_Performance_Master_Cleaned .hyper / .csv	Production-ready dashboard source